

Linear Stability Theory of the Compressible Flat-Plate Boundary Layer

The laminar mean flow, the disturbance eigenvalue problem, its spectral numerical solution, and validation against published benchmarks

Technical documentation of our LST solver

July 3, 2026

Contents

1	Overview	3
2	The mean boundary-layer profile	5
2.1	Governing equations and non-dimensionalisation	5
2.2	The self-similar base flow	6
2.3	Transformation to physical coordinates	7
3	The linearised disturbance equations	9
3.1	Linearisation and the normal-mode ansatz	9
3.2	The four disturbance equations	10
3.3	Operator form and the discretised derivative	11
3.4	Physical character of the instability modes	14
3.5	Assumptions and range of validity	15
4	Numerical methods: the spectral eigenvalue solver	17
4.1	The numerical task	17
4.2	Assembling the operator and its boundary conditions	17
4.3	The temporal problem: a generalised eigenvalue problem	20
4.4	The spatial problem: a quadratic eigenvalue problem	21
4.5	Identifying the physical mode	23
4.6	Worked example: one temporal solve	24
4.7	Neutral curves	25
4.8	N -factor and transition	26
5	Validation and verification	27
5.1	Summary of all cases	27
5.2	Incompressible core	28
5.2.1	Orszag (1971): plane-Poiseuille Orr–Sommerfeld eigenvalue	28
5.2.2	Orszag (1971) Table 5: the full Orr–Sommerfeld spectrum	28
5.3	Compressible eigenvalues	29
5.3.1	Malik (1990): compressible eigenvalues	29
5.3.2	Malik (1990) Fig. 4: M10 second-mode eigenfunction	30

5.3.3	Balakumar & Malik (1992): discrete/continuous spectrum topology	32
5.4	Second mode: curves and families	33
5.4.1	Mack (1984) Fig. 10.6: second-mode growth vs. Reynolds number	33
5.4.2	Egorov et al. (2006): Mach-6 flat-plate second mode	34
5.4.3	Ma & Zhong (2003) Fig. 15: M4.5 second-mode neutral curve	34
5.5	First mode: curves and families	36
5.5.1	Mack (1984) Fig. 10.1: first-mode neutral loop	36
5.5.2	Mack (1984) Fig. 10.3: oblique first-mode growth vs. R	37
5.5.3	Mack (1984) Fig. 10.4: oblique first mode at high Mach	38
5.6	Mixed first and second mode: the Özgen–Kırçalı neutral-curve family	39

1 Overview

Linear stability theory (LST) determines whether an infinitesimal wavelike disturbance superposed on a steady laminar boundary layer amplifies or decays as it propagates [1]. Because the disturbance is infinitesimal, its governing equations are linear; because their coefficients depend, under the parallel-flow assumption, only on the wall-normal coordinate y , the normal-mode substitution $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ removes x and t entirely, leaving an eigenvalue problem for the shape $\hat{q}(y)$. Solved repeatedly across parameter space, that eigenvalue problem yields the growth rate of every candidate instability wave, and from growth rates everything else follows: neutral curves, amplification envelopes, and the e^N transition estimate. This document is the complete technical record of that computation as implemented in our LST solver; Figure 1 shows the computation whole, and with it the layout of the document.

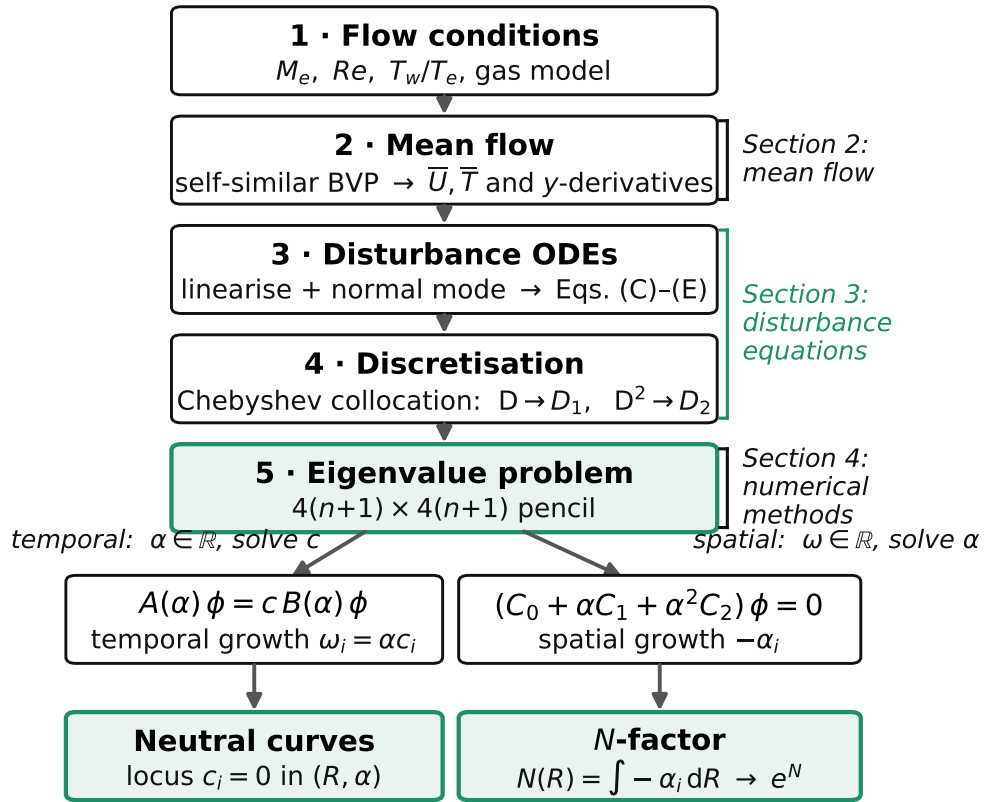


Figure 1: The complete LST computation and the layout of this document. Stages 1–2 are the subject of Section 2, stages 3–4 of Section 3, and stage 5, with the temporal/spatial fork and the two products beneath it, of Section 4; Section 5 validates the chain against published benchmarks.

The five stages compose as follows. The flow conditions (stage 1) fix a self-similar boundary-value problem whose solution (stage 2) is the laminar mean flow: the profiles $\bar{U}(y)$ and $\bar{T}(y)$ and their wall-normal derivatives, which enter every later stage as known coefficients. Linearising the Navier–Stokes equations about that mean and substituting the normal-mode ansatz (stage 3) produces four coupled disturbance ODEs, collected into the operator pencil $\mathbf{L}\hat{q} = c\mathbf{M}\hat{q}$; replacing the wall-normal derivative by Chebyshev collocation matrices (stage 4) turns the pencil into the finite generalised eigenvalue problem $A(\alpha)\phi = cB(\alpha)\phi$, the computational atom of the framework.

Its solution (stage 5) is posed in two closures: the temporal problem takes a real wavenumber α and returns the complex phase speed c , with growth rate $\omega_i = \alpha c_i$; the spatial problem takes a real frequency ω and returns the complex wavenumber α , with growth rate $\sigma = -\alpha_i$. Swept over parameter space, these solves yield the two products at the foot of the figure: the neutral curve, the locus of zero growth in the (R, α) plane, and the N -factor, the accumulated amplification $\ln[A(R)/A(R_0)]$ whose empirical threshold $N \approx 9$ –11 correlates transition [2, 3].

Section 5 closes the document by auditing this chain against published benchmarks at their exact stated conditions: the incompressible Orr–Sommerfeld spectrum of Orszag [4], the compressible eigenvalues of Malik [5], the growth-rate and neutral-curve families of Mack [1], and the mixed-mode neutral-curve family of Özgen & Kırçalı [6], among others. An independent shooting solver, used only to cross-check the collocation results, is documented separately and is not developed here.

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. A prime on a mean quantity denotes $D \equiv d/dy$ (in the similarity development of Section 2, $d/d\eta$, as stated there). The continuous amplitude vector is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$; its discrete, grid-sampled form is ϕ .

Base flow & transport		Disturbance & eigenvalue	
$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temperature, density	$\hat{q}; \phi$	amplitude shape (continuous; discrete)
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
$C=\bar{\mu}/\bar{T}$	Chapman–Rubesin group	α, ω	streamwise wavenumber, frequency
η	similarity coordinate, Eq. (14)	$c=\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\nu_R=1/(Re\bar{\rho})$	viscous prefactor	$\omega_i=\alpha c_i; \sigma=-\alpha_i$	temporal; spatial growth rate
$\mathcal{C}_\kappa, \mathcal{C}_d, \mathcal{L}_c$	conduction/dissipation groups	M_{rel}	relative Mach number
Scales & parameters		Discretisation	
M_e, Re, Pr, γ	Mach, Reynolds, Prandtl, c_p/c_v	D_1, D_2	differentiation matrices
R_g, a_e	gas constant, edge sound speed	$N; n=N+1$	collocation order; grid size
$L^*=\sqrt{\nu_e x/U_e}$	Mack length, Eq. (7)	A, B	temporal operator pair
$R=\sqrt{Re_x}; F$	Mack Reynolds number; reduced frequency	C_0, C_1, C_2	spatial operator blocks
$\alpha_L=\alpha L^*; \delta^*, \theta$	scaled wavenumber; displ./momentum thickness		

2 The mean boundary-layer profile

This section specifies how our LST solver computes the steady, laminar, compressible flat-plate boundary layer whose profiles $\bar{U}$, $\bar{T}$, $\bar{\rho}$, $\bar{\mu}$ and their wall-normal derivatives are the coefficients of the linear stability problem. Every equation, boundary condition, coefficient, and default value is stated exactly as implemented.

2.1 Governing equations and non-dimensionalisation

The flow is governed by the compressible Navier–Stokes equations, expressing conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas [7]:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (\text{mass}) \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (\text{momentum}) \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (\kappa \nabla T) + \Phi, \quad (\text{energy}) \quad (3)$$

$$p = \rho R_g T, \quad (\text{state}) \quad (4)$$

where $\mathbf{u} = (u, v)$ is the velocity, ρ the density, T the temperature, p the pressure, μ the viscosity, κ the conductivity, c_p the specific heat, R_g the specific gas constant (plain R is reserved for the Mack Reynolds number, Eq. (8)), and $D/Dt = \partial_t + \mathbf{u} \cdot \nabla$ the material derivative. The remaining terms are the Newtonian viscous stress tensor and the dissipation function,

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{u} + \nabla \mathbf{u}^\top - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right), \quad \Phi = \boldsymbol{\tau} : \nabla \mathbf{u}. \quad (5)$$

Their closed forms are not required in what follows; it suffices that both are linear in the velocity gradients. Equations (1)–(4) are applied twice: once to obtain the steady mean flow below, and once in linearised form about that mean flow to obtain the disturbance equations of Section 3.

Non-dimensionalisation and default gas constants. Velocities are scaled by the edge speed U_e , temperature by T_e , density by ρ_e , viscosity by μ_e , and pressure by $\rho_e U_e^2$. Because the gas is perfect and the pressure is nearly uniform across the layer, $\bar{\rho} = 1/\bar{T}$ in these units. Three dimensionless groups govern the problem,

$$M_e = \frac{U_e}{a_e}, \quad a_e = \sqrt{\gamma R_g T_e}, \quad Re = \frac{\rho_e U_e L_{\text{ref}}}{\mu_e}, \quad Pr = \frac{\mu_e c_p}{\kappa_e}, \quad (6)$$

where a_e is the edge speed of sound. The default gas constants are the standard air values $\gamma = 1.4$, $Pr = 0.72$, Sutherland constant $S = 110.4$ K [8], and power-law exponent $\omega_\mu \approx 0.7$ when the power-law viscosity model is selected.

The Mack length and Reynolds scales. A flat-plate boundary layer grows as $\sqrt{x}$, so the natural local length scale at station x , following Mack [1], is

$$L^* = \sqrt{\frac{\nu_e x}{U_e}}, \quad \nu_e = \frac{\mu_e}{\rho_e}. \quad (7)$$

With $L_{\text{ref}} = L^*$ the Reynolds number becomes the square root of the conventional one, and a fixed physical frequency f maps to a reduced frequency F ; both enter the neutral-curve and N -factor computations:

$$R = \frac{U_e L^*}{\nu_e} = \sqrt{Re_x}, \quad Re_x = \frac{U_e x}{\nu_e}; \quad F = \frac{2\pi f \nu_e}{U_e^2}; \quad \alpha_L = \alpha L^*, \quad (8)$$

where α is the streamwise wavenumber. R is the abscissa of every neutral curve and α_L the ordinate. The solver itself is dimensionless; the conversions above are pure postprocessing.

2.2 The self-similar base flow

The mean flow is obtained from a boundary-value problem, not an eigenvalue problem: the data are prescribed at both ends of the domain ($y = 0$ and $y \rightarrow \infty$) and the solution is unique. Its unknowns are the steady laminar profiles $\bar{U}$, $\bar{T}$, $\bar{\rho}$, $\bar{\mu}$ and their wall-normal derivatives, which subsequently enter the linearised disturbance equations as known coefficients.

For a flat plate the mean flow is *self-similar*: profiles at different streamwise stations collapse onto a single curve in the stretched coordinate η . Introducing the dimensionless stream function $f(\eta)$ with $f' = \bar{U} = u/U_e$ (in this section a prime denotes $d/d\eta$), the momentum and energy equations (2)–(3) reduce under the boundary-layer approximation to the classical pair of coupled ordinary differential equations [7, 6],

$$(C f'')' + f f'' = 0, \quad (9)$$

$$(B \bar{T}')' + f \bar{T}' + (\gamma - 1) M_e^2 C (f'')^2 = 0, \quad (10)$$

with the transport groups

$$C = \frac{\mu/\mu_e}{T/T_e}, \quad K = \frac{\kappa}{c_p \mu_e}, \quad B = \frac{K}{\bar{T}} \left(= \frac{C}{Pr} \text{ at constant } Pr \right), \quad (11)$$

where C is the Chapman–Rubesin parameter [9]. The final term of (10) represents aerodynamic heating: it elevates the near-wall temperature at high Mach number and couples the energy equation to the velocity field. The conductivity model is $\kappa = \mu c_p / Pr$, so that $\bar{\kappa} = \bar{\mu} / Pr$ at constant Pr ; the transport derivatives $d\mu/dT$, $d\kappa/dT$ are evaluated analytically from the selected viscosity law.

Boundary conditions. Equations (9)–(10) are closed by

$$\underbrace{f(0) = 0, \quad f'(0) = 0}_{\text{no slip}}, \quad \underbrace{f'(\infty) = 1, \quad \bar{T}(\infty) = 1}_{\text{match the edge}}, \quad \begin{cases} \bar{T}'(0) = 0 & (\text{adiabatic wall}) \\ \bar{T}(0) = T_w/T_e & (\text{isothermal wall}) \end{cases} \quad (12)$$

For an adiabatic wall the recovery temperature is the standard result $T_{aw}/T_e = 1 + \frac{1}{2}(\gamma - 1)\sqrt{Pr} M_e^2$ [7].

Figure 2 shows the computed profiles for a representative hypersonic case: $M_e = 6$, edge temperature $T_e = 54$ K, and an isothermal wall at $T_w = 300$ K, i.e. $T_w/T_e = 5.56$. Since the adiabatic recovery temperature at these conditions is $T_{aw} \approx 7.1 T_e$, the wall is cooled. The velocity rises monotonically to its edge value, while viscous heating produces an interior temperature maximum, $\bar{T} \approx 5.7 T_e$, just above the cooled wall. The dotted line marks the outermost generalised inflection point, $(\bar{\rho} \bar{U}')' = 0$, the seat of the inviscid instability identified by Lees & Lin [10].

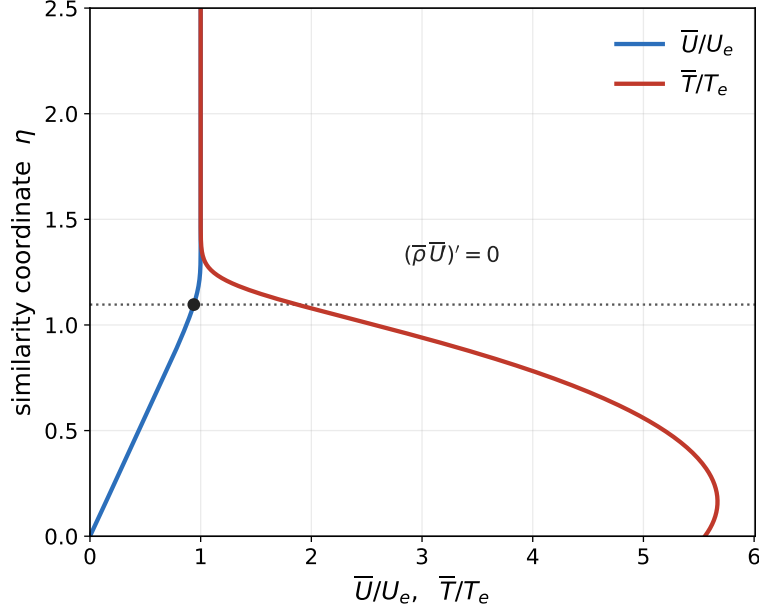


Figure 2: Computed compressible mean flow in the similarity coordinate η : $M_e = 6$, $T_e = 54$ K, isothermal wall at $T_w = 300$ K ($T_w/T_e = 5.56$, cooled relative to $T_{aw} \approx 7.1 T_e$). Viscous heating produces an interior temperature maximum $\approx 5.7 T_e$ just above the wall. The dotted line marks the outermost generalised inflection point $(\bar{\rho} \bar{U})' = 0$ [10]. These profiles and their wall-normal derivatives are the known coefficients of the linearised disturbance equations; the transformation to physical coordinates is given by Eqs. (14) and (15).

2.3 Transformation to physical coordinates

The similarity solution is computed once, in η ; all case-specific dimensional information enters only through the transformation back to the physical wall-normal coordinate. Because the near-wall gas is hot ($\bar{T} > 1$) and correspondingly rarefied ($\bar{\rho} = 1/\bar{T}$), equal increments of η span unequal increments of physical distance. The map follows directly from the temperature field. In the units of Section 2.1 the differential relation is

$$\frac{dy}{d\eta} = \sqrt{2\bar{T}(\eta)} \quad (y \text{ in units of } L^*), \quad (13)$$

whose integral gives the physical coordinate at station x ,

$$\frac{y}{L^*} = \sqrt{2} \int_0^\eta \bar{T}(\eta') d\eta'. \quad (14)$$

Substituting the Mack length (7) yields the fully dimensional, case-specific form

$$y^*(\eta; x^*) = \sqrt{\frac{2\nu_e x^*}{U_e}} \int_0^\eta \bar{T}(\eta') d\eta' \quad (15)$$

so that a single similarity solution generates the entire family of dimensional profiles, parameterised by the streamwise station x^* and the edge state (U_e, ν_e) . The map (14) is strictly monotone, and its inverse assigns a unique η to any physical height y^* . Since $\bar{T} > 1$ near the wall, the transform outpaces the linear reference $y/L^* = \sqrt{2}\eta$ there: a uniform η mesh maps to a physical mesh

stretched into the hot near-wall region, which is why η , rather than y , is the natural computational coordinate. Wall-normal derivatives convert by the chain rule implied by (13),

$$\frac{d}{dy} = \frac{1}{\sqrt{2}\overline{T}(\eta)} \frac{d}{d\eta} \quad (y \text{ in units of } L^*). \quad (16)$$

The integral thicknesses follow from the same transform,

$$\frac{\delta^*}{L^*} = \sqrt{2} \int_0^\infty (\overline{T} - \overline{U}) d\eta, \quad \frac{\theta}{L^*} = \sqrt{2} \int_0^\infty \overline{U} (1 - \overline{U}) d\eta, \quad (17)$$

where the displacement thickness δ^* measures the outward displacement of the outer flow by the layer and the momentum thickness θ its momentum deficit. A profile may be stored on a y/δ^* or a y/L^* grid; the constant δ^*/L^* converts between them, with $D \mapsto (\delta^*/L^*)^{-1}D$.

The profiles $\overline{U}, \overline{T}, \overline{\rho}, \overline{\mu}$ and their wall-normal derivatives assembled here are the coefficients of the linearised disturbance equations derived next.

3 The linearised disturbance equations

3.1 Linearisation and the normal-mode ansatz

The disturbance obeys the same compressible Navier–Stokes equations (1)–(4) that produced the mean flow of Section 2, with the stress tensor and dissipation function (5) and the non-dimensionalisation of Section 2.1; they are now applied a second time, in linearised form, about that mean. The reduction from four partial differential equations to four ordinary differential equations in y proceeds in five steps.

Step 1: decomposition into mean and disturbance. Each field is split as

$$u = \bar{U}(y) + u', \quad v = v', \quad T = \bar{T}(y) + T', \quad p = \bar{p} + p', \quad \rho = \bar{\rho}(y) + \rho', \quad |\cdot| \ll |\bar{\cdot}|, \quad (18)$$

with $\bar{v} = 0$ for a parallel mean flow. The collection $q' = (u', v', T', p', \rho')$ is the disturbance.

Step 2: linearisation. Substituting (18) into (1)–(4), the mean satisfies the equations by itself and cancels; terms quadratic in the primed quantities are dropped. The remaining equations are linear in q' with coefficients constructed from the known mean profiles.

Step 3: parallel-flow assumption. The boundary layer grows slowly, so locally $\bar{q} = \bar{q}(y)$ only and the coefficients depend on y alone. This neglects the $O(1/R)$ streamwise growth of the mean flow and of the eigenfunction, which the parabolised stability equations retain [11] (cf. Section 3.5).

Step 4: normal-mode ansatz. The disturbance is represented as a single travelling wave,

$$q'(x, y, t) = \hat{q}(y) e^{i(\alpha x - \omega t)}, \quad \hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top, \quad (19)$$

with α the streamwise wavenumber, ω the frequency, and $c = \omega/\alpha$ the complex phase speed. Substitution replaces $\partial_x \mapsto i\alpha$, $\partial_t \mapsto -i\omega$, and $\partial_y \mapsto D$, collapsing the linear PDEs to coupled ODEs in y (Figure 3). The spanwise wavenumber is set to zero ($\beta = 0$); a general oblique wave $\propto e^{i(\alpha x + \beta z - \omega t)}$ propagates at angle $\psi = \arctan(\beta/\alpha)$.

Step 5: elimination of density. The linearised equation of state gives

$$\frac{\rho'}{\bar{\rho}} = \gamma M_e^2 \hat{p} - \frac{\hat{T}}{\bar{T}}, \quad (20)$$

which removes ρ' from the system. Four amplitude fields remain, $\hat{u}, \hat{v}, \hat{T}, \hat{p}$, with $\hat{p}$ scaled by $\rho_e U_e^2$ (so that $\gamma M_e^2 \hat{p} = p'/p_e$).

A worked linearisation. To make the mechanics explicit, consider mass conservation (1), written out as $\partial_t \rho + \partial_x(\rho u) + \partial_y(\rho v) = 0$. Inserting the decomposition (18) and expanding every product sorts the equation into three groups:

$$\underbrace{\partial_t \bar{\rho} + \partial_x(\bar{\rho} \bar{U})}_{\text{mean: vanishes}} + \underbrace{\partial_t \rho' + \bar{U} \partial_x \rho' + \bar{\rho} \partial_x u' + \partial_y(\bar{\rho} v')}_{\text{linear in } q': \text{ retained}} + \underbrace{\partial_x(\rho' u') + \partial_y(\rho' v')}_{O(|q'|^2): \text{ dropped}} = 0. \quad (21)$$

The mean group vanishes identically: the base flow is steady, $\partial_t \bar{\rho} = 0$, and parallel, so $\bar{\rho}$ and $\bar{U}$ depend on y alone and $\partial_x(\bar{\rho} \bar{U}) = 0$; in general the mean satisfies its own conservation equations

and cancels from the disturbance system. The same parallel-flow property has already simplified the linear group, since $\partial_x(\bar{\rho} u') = \bar{\rho} \partial_x u'$ and $\partial_x(\rho' \bar{U}) = \bar{U} \partial_x \rho'$. The quadratic group is negligible by the smallness assumption in (18). What survives is the linear equation

$$\partial_t \rho' + \bar{U} \partial_x \rho' + \bar{\rho} \partial_x u' + \partial_y(\bar{\rho} v') = 0. \quad (22)$$

Applying the ansatz (19) replaces $\partial_t \rightarrow -i\alpha c$, $\partial_x \rightarrow i\alpha$, $\partial_y \rightarrow D$ and each primed field by its amplitude, so that the first two terms combine into the convective factor:

$$i\alpha(\bar{U} - c) \hat{\rho} + i\alpha \bar{\rho} \hat{u} + \bar{\rho} D \hat{v} + \bar{\rho}' \hat{v} = 0. \quad (23)$$

Dividing by $\bar{\rho}$, substituting $\hat{\rho}/\bar{\rho} = \gamma M_e^2 \hat{p} - \hat{T}/\bar{T}$ from (20), and using $\bar{\rho}'/\bar{\rho} = -\bar{T}'/\bar{T}$ (from $\bar{\rho} = 1/\bar{T}$) yields exactly Eq. (24) below. The remaining three equations follow by the identical procedure applied to (2)–(3); the algebra is mechanical but lengthy, and only the result is printed.

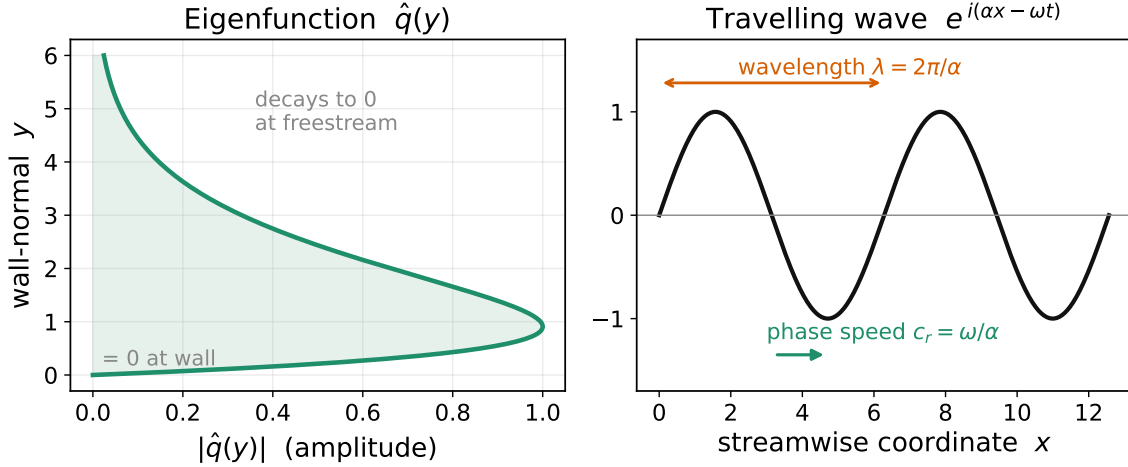


Figure 3: The ansatz (19) factorises a disturbance $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ into a wall-normal *shape* $\hat{q}(y)$ (left; the eigenfunction, zero at the wall and decaying at the edge) and a travelling wave in x (right; wavelength $2\pi/\alpha$, phase speed $c_r = \text{Re}(c)$).

3.2 The four disturbance equations

The linearised system consists of four coupled ODEs in y for $\hat{u}, \hat{v}, \hat{T}, \hat{p}$: continuity (with the equation of state folded in through (20)), streamwise and wall-normal momentum, and energy. Fixing one of α or ω real leaves the other (through c or α) as the eigenvalue. Every overbarred quantity is a known base-flow coefficient; all transport derivatives ($d\mu/dT$, $d\kappa/dT$, ...) are evaluated at the local mean temperature $\bar{T}(y)$ and are therefore known functions of y . The form below follows Özgen & Kırcah [6]; it is equivalent to the canonical compressible stability equations of Mack [1] and Malik [5] restricted to two dimensions. Momentum and energy are divided by $\bar{\rho}$, and the viscous prefactor is written $\nu_R \equiv 1/(Re \bar{\rho})$; every $\bar{\rho}$ is the base-flow density $\bar{\rho}(y) = 1/\bar{T}(y)$, uniformly inside ν_R , $\mathcal{C}_\kappa$, and $\mathcal{C}_d$.

(C) Continuity + equation of state.

$$i\alpha \hat{u} + D \hat{v} - \frac{\bar{T}'}{\bar{T}} \hat{v} + i\alpha(\bar{U} - c) \left(\gamma M_e^2 \hat{p} - \frac{\hat{T}}{\bar{T}} \right) = 0. \quad (24)$$

(X) Streamwise momentum.

$$\begin{aligned} i\alpha(\bar{U} - c)\hat{u} + \bar{U}'\hat{v} + i\alpha\bar{\rho}^{-1}\hat{p} = \nu_R \Big[\bar{\mu}(D^2 - \frac{4}{3}\alpha^2)\hat{u} + \bar{\mu}'D\hat{u} + i\alpha(\frac{1}{3}\bar{\mu}D + \bar{\mu}')\hat{v} \\ + \frac{d\bar{\mu}}{dT}(\bar{U}'' + \bar{U}'D)\hat{T} + \frac{d^2\bar{\mu}}{dT^2}\bar{U}'\hat{T}\hat{T} \Big]. \end{aligned} \quad (25)$$

(Y) Wall-normal momentum.

$$i\alpha(\bar{U} - c)\hat{v} + \bar{\rho}^{-1}D\hat{p} = \nu_R \Big[\bar{\mu}(\frac{4}{3}D^2 - \alpha^2)\hat{v} + \frac{4}{3}\bar{\mu}'D\hat{v} + i\alpha(\frac{1}{3}\bar{\mu}D - \frac{2}{3}\bar{\mu}')\hat{u} + i\alpha\frac{d\bar{\mu}}{dT}\bar{U}'\hat{T} \Big]. \quad (26)$$

(E) Energy (temperature).

$$i\alpha(\bar{U} - c)\hat{T} + \bar{T}'\hat{v} + (\gamma - 1)\bar{\rho}^{-1}(i\alpha\hat{u} + D\hat{v}) = \mathcal{C}_\kappa\mathcal{L}_c\hat{T} + \mathcal{C}_d \Big[2\bar{\mu}\bar{U}'(D\hat{u} + i\alpha\hat{v}) + \frac{d\bar{\mu}}{dT}(\bar{U}')^2\hat{T} \Big], \quad (27)$$

$$\mathcal{L}_c = D^2 - \alpha^2 + \frac{1}{\bar{\kappa}}\frac{d\bar{\kappa}}{dT}(2\bar{T}'D + \bar{T}'') + \frac{1}{\bar{\kappa}}\frac{d^2\bar{\kappa}}{dT^2}(\bar{T}')^2, \quad \mathcal{C}_\kappa = \frac{\gamma\bar{\mu}}{Pr Re \bar{\rho}}, \quad \mathcal{C}_d = \frac{\gamma(\gamma - 1)M_e^2}{Re \bar{\rho}}. \quad (28)$$

The Prandtl number in $\mathcal{C}_\kappa$ is taken as the local value when the selected transport model supplies a temperature-dependent Pr ; for the constant- Pr default the two coincide.

Structure of the equations. The following observations locate the physics in (24)–(27):

- $i\alpha(\bar{U} - c)$ is convection by the mean flow relative to the wave; it vanishes at the *critical layer* $\bar{U} = c_r$, where the inviscid form of the equations becomes singular, which is why that layer governs much of the instability physics [1].
- D, D^2 are wall-normal derivatives; the $-\alpha^2$ terms arise from $\partial_x^2 \mapsto -\alpha^2$.
- The $\nu_R = 1/(Re\bar{\rho})$ terms are viscous; the factors $\frac{4}{3}, \frac{1}{3}$ correspond to the Stokes hypothesis for the bulk viscosity.

3.3 Operator form and the discretised derivative

Equations (24)–(27) are linear both in the amplitude vector $\hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top$ and in the eigenvalue, and every derivative they contain is of at most second order. They can therefore be collected into the single operator statement

$$L(\alpha; y, D) \hat{q} = c M(\alpha; y) \hat{q}, \quad L = A_2(y) D^2 + A_1(\alpha; y) D + A_0(\alpha; y), \quad (29)$$

with rows ordered (C, X, Y, E) and columns ordered $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$. Two structural facts make (29) particularly convenient for what follows.

First, the eigenvalue c enters (24)–(27) *only* through the convective factor $i\alpha(\bar{U} - c)$. The right-hand operator M is therefore purely algebraic, containing no derivatives, and can be written down at once:

$$M(\alpha; y) = i\alpha \begin{pmatrix} 0 & 0 & -1/\bar{T} & \gamma M_e^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (30)$$

Second, only three terms in the entire system carry D^2 : the viscous diffusion of $\hat{u}$ and $\hat{v}$ in (25)–(26) and the heat conduction of $\hat{T}$ in (27). The second-order coefficient matrix is therefore

sparse:

$$\mathbf{A}_2(y) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\nu_R \bar{\mu} & 0 & 0 & 0 \\ 0 & -\frac{4}{3} \nu_R \bar{\mu} & 0 & 0 \\ 0 & 0 & -\mathcal{C}_\kappa & 0 \end{pmatrix}, \quad (31)$$

where the signs follow from moving the right-hand sides of (25)–(27) to the left. The pressure amplitude $\hat{p}$ is never differentiated twice, so its column in $\mathbf{A}_2$ is empty, reflecting the absence of a pressure diffusion mechanism.

The first- and zeroth-order matrices are read off from (24)–(27) in the same manner. For the energy row it is convenient to split the conduction operator (28) by derivative order,

$$\mathcal{L}_c = \mathbf{D}^2 + \ell_1 \mathbf{D} + \ell_0, \quad \ell_1 = \frac{2\bar{T}'}{\bar{\kappa}} \frac{d\kappa}{dT}, \quad \ell_0 = -\alpha^2 + \frac{\bar{T}''}{\bar{\kappa}} \frac{d\kappa}{dT} + \frac{(\bar{T}')^2}{\bar{\kappa}} \frac{d^2\kappa}{dT^2}. \quad (32)$$

The first-order matrix collects every term carrying a single $\mathbf{D}$:

$$\mathbf{A}_1(\alpha; y) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\nu_R \bar{\mu}' & -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\nu_R \frac{d\mu}{dT} \bar{U}' & 0 \\ -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\frac{4}{3} \nu_R \bar{\mu}' & 0 & \bar{\rho}^{-1} \\ -2\mathcal{C}_d \bar{\mu} \bar{U}' & (\gamma - 1) \bar{\rho}^{-1} & -\mathcal{C}_\kappa \ell_1 & 0 \end{pmatrix}, \quad (33)$$

and the zeroth-order matrix collects everything algebraic that does not multiply c :

$$\mathbf{A}_0(\alpha; y) = \begin{pmatrix} i\alpha & -\frac{\bar{T}'}{\bar{T}} & -\frac{i\alpha \bar{U}}{\bar{T}} & i\alpha \gamma M_e^2 \bar{U} \\ i\alpha \bar{U} + \frac{4}{3} \alpha^2 \nu_R \bar{\mu} & \bar{U}' - i\alpha \nu_R \bar{\mu}' & -\nu_R \left(\frac{d\mu}{dT} \bar{U}'' + \frac{d^2\mu}{dT^2} \bar{U}' \bar{T}' \right) & i\alpha \bar{\rho}^{-1} \\ \frac{2i\alpha}{3} \nu_R \bar{\mu}' & i\alpha \bar{U} + \alpha^2 \nu_R \bar{\mu} & -i\alpha \nu_R \frac{d\mu}{dT} \bar{U}' & 0 \\ i\alpha (\gamma - 1) \bar{\rho}^{-1} & \bar{T}' - 2i\alpha \mathcal{C}_d \bar{\mu} \bar{U}' & i\alpha \bar{U} - \mathcal{C}_\kappa \ell_0 - \mathcal{C}_d \frac{d\mu}{dT} (\bar{U}')^2 & 0 \end{pmatrix}. \quad (34)$$

Temporal and spatial problems. Equation (29) supports the two standard closures. Fixing $\alpha \in \mathbb{R}$ leaves c as the eigenvalue of a *linear* operator pencil; this is the temporal problem. Fixing $\omega \in \mathbb{R}$ instead and regrouping the same equations by powers of α gives

$$(\mathbf{C}_0(\omega; y, \mathbf{D}) + \alpha \mathbf{C}_1(\omega; y, \mathbf{D}) + \alpha^2 \mathbf{C}_2(y)) \hat{q} = 0, \quad (35)$$

a *quadratic* pencil in α ; this is the spatial problem, reduced to linear form by companion linearisation at the numerical stage.

Chebyshev collocation. Up to this point $\mathbf{D}$ has denoted the continuous derivative d/dy acting on smooth functions. To make (29) computable, each amplitude is represented by its values at a finite set of grid points and $\mathbf{D}$ by a matrix acting on those values. For smooth functions a polynomial interpolant through Chebyshev points converges exponentially with the polynomial degree [4, 12], so few points yield many digits; this makes spectral collocation the natural discretisation for boundary-layer eigenvalue problems [5]. The Chebyshev–Gauss–Lobatto nodes on $[-1, 1]$ are

$$\xi_j = \cos\left(\frac{\pi j}{n}\right), \quad j = 0, 1, \dots, n, \quad (36)$$

clustering near the ends $\xi = \pm 1$; the ordering runs from $\xi_0 = +1$ down to $\xi_n = -1$.

The differentiation matrix. Exactly one polynomial of degree $\leq n$ passes through the sampled values $\{f(\xi_j)\}$; differentiating that interpolant and evaluating the result at the nodes is a linear operation on the values, $f'(\xi_j) = \sum_k (D_\xi)_{jk} f(\xi_k)$, with entries [13, 12]

$$(D_\xi)_{jk} = \frac{c_j}{c_k} \frac{(-1)^{j+k}}{\xi_j - \xi_k} \quad (j \neq k), \quad (D_\xi)_{jj} = -\sum_{k \neq j} (D_\xi)_{jk}, \quad c_0 = c_n = 2, \quad c_j = 1 \text{ otherwise.} \quad (37)$$

The diagonal is fixed by requiring each row to sum to zero, since a constant function must have zero derivative, and $c_0 = c_n = 2$ accounts for the doubled weight of the two endpoint nodes. As an illustration, the smallest useful case $n = 2$ (three nodes $\xi = (1, 0, -1)$) gives

$$D_\xi = \begin{pmatrix} \frac{3}{2} & -2 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 2 & -\frac{3}{2} \end{pmatrix}, \quad D_\xi^2 = D_\xi D_\xi = \begin{pmatrix} 1 & -2 & 1 \\ 1 & -2 & 1 \\ 1 & -2 & 1 \end{pmatrix}. \quad (38)$$

The middle row of D_ξ is the familiar central difference $\frac{1}{2}(f_0 - f_2)$, and the rows of D_ξ^2 are identical because the second derivative of the parabola through three points is constant. Both matrices are exact on polynomials of degree $\leq n$. Production runs use far more nodes, typically $n \sim 100$ – 200 , so that the eigenfunctions are resolved to several digits. At any order the matrices are dense, because every node's derivative uses every node's value, and this density is inherited by the assembled stability matrices below.

Mapping to the physical layer. The flow occupies $y \in [0, y_{\max}]$ with resolution needed at the wall, where the eigenfunctions vary fastest. The algebraic map of Malik [5],

$$y(\xi) = L \frac{1 + \xi}{1 - \xi + 2L/y_{\max}}, \quad L = \frac{y_{\max}}{6}, \quad (39)$$

sends $\xi = +1$ (node $j=0$) to $y = y_{\max}$, the freestream, and $\xi = -1$ (node $j=n$) to $y = 0$, the wall, placing roughly half the nodes below $y \approx L$. By the chain rule, with $\xi_y = 1/y_\xi$ and $\xi_{yy} = -y_{\xi\xi}/y_\xi^3$, the differentiation matrices on the physical grid $\{y_j\}_{j=0}^n$ are

$$D_1 = \text{diag}(\xi_y) D_\xi, \quad D_2 = \text{diag}(\xi_y^2) D_\xi^2 + \text{diag}(\xi_{yy}) D_\xi, \quad (40)$$

where the chain rule acts on the computational square D_ξ^2 : the physical D_2 is *not* $D_1 D_1$.

The discrete eigenvalue problem. Numerical solution now amounts to the substitution

$$D \longrightarrow D_1, \quad D^2 \longrightarrow D_2, \quad D_1, D_2 \in \mathbb{R}^{(n+1) \times (n+1)}. \quad (41)$$

Under (41) each mean-flow coefficient becomes a diagonal matrix, each entry of A_2, A_1, A_0, M becomes an $(n+1) \times (n+1)$ block, and the four sampled amplitudes are stacked into a single vector $\phi = (\hat{u}; \hat{v}; \hat{T}; \hat{p}) \in \mathbb{C}^{4(n+1)}$. The operator statement (29) then becomes the finite-dimensional generalised eigenvalue problem

$$\boxed{A(\alpha) \phi = c B(\alpha) \phi} \quad A, B \in \mathbb{C}^{4(n+1) \times 4(n+1)}. \quad (42)$$

Both matrices inherit the 4×4 block structure of the operator form. With i, j over the rows (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$, every block follows one assembly rule,

$$A_{ij}(\alpha) = \Lambda([A_2]_{ij}) D_2 + \Lambda([A_1]_{ij}) D_1 + \Lambda([A_0]_{ij}), \quad B_{ij}(\alpha) = \Lambda([M]_{ij}), \quad (43)$$

where $\Lambda(f) = \text{diag}(f(y_0), \dots, f(y_n))$ places a coefficient on the grid. Three ingredients therefore build the whole problem: the differentiation matrices D_1, D_2 , diagonal samples of the mean-flow profiles, and the scalars α, M_e, Re .

One solve: input and output. A single solve of (42) takes one point of parameter space, the mean-flow profiles at (M_e, Re) and one real wavenumber α , and returns the full discrete spectrum of $4(n+1)$ eigenpairs (c, ϕ) , each a candidate normal mode:

- c_r is the phase speed as a fraction of U_e ;
- c_i sets the temporal growth rate $\omega_i = \alpha c_i$: the wave amplifies for $c_i > 0$, decays for $c_i < 0$, and is neutral at $c_i = 0$;
- the four segments of ϕ are the wave’s velocity, temperature, and pressure amplitude profiles across the layer (a wall-peaked $|\hat{p}|$, for instance, signals a second mode).

Only a few of these eigenpairs are physical *discrete* modes; the rest approximate the continuous spectra or are numerical artifacts. The boundary-condition rows carry no eigenvalue and leave zero rows in B , so B is singular and the pencil is solved in generalised (QZ) form, which produces spurious infinite eigenvalues that are discarded.

Everything the framework produces is built from repeated solves of (42): one solve interrogates one point (M_e, Re, α) ; a sweep traces the growth rate across the (R, α) plane; the zero level set $c_i = 0$ of that sweep is the neutral curve; and the integral of the growth rate along frequency lines is the N -factor. Solving the pencil and identifying the physical modes within the spectrum is the remaining numerical task.

3.4 Physical character of the instability modes

The disturbance equations admit discrete eigenmodes; a temporal solve returns a complex phase speed such as $c = 0.930 + 0.0200i$. This section summarises the physics behind the standard mode labels, so that such a number identifies a definite wave.

First (Tollmien–Schlichting) mode. At low speed the first mode is a viscous instability ($c_r \approx 0.3$ – 0.6): viscosity introduces a phase lag between u' and v' near the wall, so the Reynolds stress $-\bar{\rho} \overline{u'v'}$ no longer averages to zero and extracts energy from the mean shear. As the Mach number rises, compressibility creates a generalised inflection point (below) and the first mode becomes increasingly inviscid and inflectional, with viscosity then often acting to stabilise; it is the dominant instability only up to $M_e \approx 4$ [1].

Second (Mack) mode. Above $M_e \approx 4$ a second, usually stronger, instability appears: an acoustic wave trapped between the wall and the relative sonic line, reflecting within a near-wall acoustic waveguide [1, 15]. This interpretation accounts for its principal properties:

- its phase speed is $c_r \approx 1 - 1/M_e$, the limiting value at which the wave is sonic relative to the edge flow, which is why the spatial-solve shift target $\alpha_0 = \omega/(1 - 1/M_e)$ is effective; the realised c_r sits somewhat above this and varies along the branch (0.93 against $1 - 1/6 = 0.83$ for a representative $M_e = 6$ mode);
- it is frequency-tuned: as in any resonator the wavelength is set by the cavity thickness, so the most-amplified wavelength is $\sim 2\delta$ and the frequency scales as $f \sim U_e/2\delta$, rising as the layer thins downstream;
- its eigenfunction has a wall-peaked pressure amplitude, the acoustic field of the waveguide.

The organising quantity: relative Mach number. Both mechanisms are organised by the relative Mach number

$$M_{\text{rel}}(y) = \frac{\bar{U}(y) - c_r}{\bar{a}(y)}, \quad \bar{a} = \frac{\sqrt{T}}{M_e}, \quad (44)$$

with $\bar{a}$ the local sound speed. The critical layer is the height at which $M_{\text{rel}} = 0$, i.e. $\bar{U} = c_r$, where the eigenfunction typically peaks and the convective factor $i\alpha(\bar{U} - c)$ in (24)–(27) vanishes; the relative sonic line is $|M_{\text{rel}}| = 1$; and the second-mode acoustic waveguide is the near-wall region in which $|M_{\text{rel}}| > 1$.

The generalised inflection point. By Rayleigh’s theorem an inviscid incompressible shear flow requires a velocity inflection point ($\bar{U}'' = 0$) for instability. Lees & Lin [10] generalised this to compressible flow: the necessary condition for the inviscid first mode is a *generalised inflection point*, $(\bar{\rho}\bar{U}')' = 0$. Aerodynamic heating places a hot, low-density layer near the wall and pushes that point up into the layer, giving compressible boundary layers a robust inviscid first-mode instability that low-speed layers lack. The second mode requires no inflection point; its existence condition is instead the relative-supersonic region $|M_{\text{rel}}| > 1$ above.

The wall-cooling reversal. Cooling the wall stabilises the first mode (it weakens the near-wall viscous mechanism) but destabilises the second (it thickens the relative-supersonic region and raises the tuned frequency) [1]. Cold-walled hypersonic vehicles are therefore second-mode dominated.

Phase and group velocity. c_r is the phase speed; energy propagates at the group velocity $c_g = d\omega_r/d\alpha_r$, and it is c_g that rigorously links temporal to spatial growth [16], the relation invoked whenever results are converted between the two formulations.

The following comparison places any computed eigenvalue at a glance:

	First (TS) mode	Second (Mack) mode
mechanism	viscous (low speed) $\rightarrow$ inviscid inflectional (high speed)	trapped acoustic wave (wall-sonic line)
phase speed c_r	0.3–0.6	$\approx 1 - 1/M_e$
dominant when	low speed, $M_e \lesssim 4$	high speed, $M_e \gtrsim 4$
frequency	broadband	tuned, $f \sim U_e/2\delta$
existence criterion	generalised inflection point [10]	relative-supersonic region, $ M_{\text{rel}} > 1$
wall cooling	stabilises	destabilises
most-unstable wave	oblique (3D)	2D ($\psi = 0$)

3.5 Assumptions and range of validity

The equations derived above are the two-dimensional, parallel-flow, modal disturbance equations. Four idealisations bound their interpretation:

1. **Parallel flow.** Setting $\bar{v} = 0$ with y -only coefficients (Step 3) neglects the $O(1/R)$ stream-wise growth of the layer, which the parabolised stability equations retain by marching the disturbance downstream [11].
2. **Two-dimensionality.** The ansatz takes $\beta = 0$ (Step 4). The most-unstable first mode is oblique ($\psi \sim 55^\circ$; Squire’s theorem does not hold under compressibility [1]), so two-dimensional first-mode growth is a lower bound; the second mode is most unstable at $\psi = 0$, so the two-dimensional result is its physical maximum.
3. **Modal analysis.** Eigenvalues describe asymptotic exponential growth; the non-normal operator also permits transient (algebraic) growth that modal analysis misses, relevant to bypass and blunt-body transition paths [14].

4. **Receptivity and e^N .** LST provides the amplification *ratio* $A(R)/A(R_0) = e^N$, not the initial amplitude $A(R_0)$; determining that amplitude is the separate receptivity problem, and the e^N method is an environment-dependent correlation rather than a first-principles criterion [3, 15]. None of these affect the equations themselves; they delimit their range of validity. A strong second mode is captured faithfully by the two-dimensional modal equations derived here, which is why they are the workhorse of hypersonic transition prediction [1, 15]. With the system (24)–(27) in operator form (29) and assembled into its discrete counterpart (42), the derivation is complete: what remains is to solve the matrix problem and identify its physical modes.

4 Numerical methods: the spectral eigenvalue solver

4.1 The numerical task

Section 3 closed with three objects: the four linearised disturbance ODEs (24)–(27), their operator form (29), and, built from the Chebyshev differentiation matrices D_1, D_2 constructed there, its discrete image: the generalised eigenvalue problem (42) on the wall-clustered collocation grid. The present section supplies everything between that statement and a trusted growth rate: the block-by-block assembly, the boundary-condition rows, the eigenvalue algorithms for the temporal and spatial closures, the identification of the physical mode within the computed spectrum, and the sweeps that turn eigenvalues into neutral curves and N -factors.

What is sought. For fixed flow parameters (M_e, Re) the disturbance equations admit non-trivial solutions only at isolated eigenvalues, and these eigenvalues are the entire objective of the computation:

- *Temporal problem:* given a real wavenumber α , find the complex phase speed c and mode shape; the temporal growth rate is $\omega_i = \alpha c_i$.
- *Spatial problem:* given a real frequency ω , find the complex wavenumber α and mode shape; the spatial growth rate is $\sigma = -\alpha_i$.

Every quantity the framework ultimately delivers is assembled from repeated eigenvalue solves: the *neutral curve* is the zero-growth locus $c_i(R, \alpha) = 0$ traced over a parameter sweep; the *N -factor* is the streamwise integral of the spatial growth rate at fixed physical frequency; and the transition estimate is the empirical correlation $N \approx 9$ –11 [2, 3]. The numerical task is therefore to evaluate eigenpairs reliably, including the rejection of spurious eigenvalues, and cheaply enough to sweep them by the thousands.

Grid indexing. Throughout this section the collocation order is N and the grid has $n = N+1$ points, so the assembled matrices are $4n \times 4n$; Section 3 wrote the same size as $4(n+1) \times 4(n+1)$, its grid index running $0, \dots, n$.

4.2 Assembling the operator and its boundary conditions

The operator pencil (29), rows ordered (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$, arrives here together with the objects that discretise it: the Chebyshev–Gauss–Lobatto nodes mapped to $y \in [0, y_{\max}]$ by the wall-clustering transform (39), node $j=0$ at the freestream and node $j=N$ at the wall, and the physical differentiation matrices D_1, D_2 of (40). Two practical choices complete the grid before assembly can begin.

Domain height and resolution. The truncation height must clear the boundary layer by a wide margin and is therefore set as a multiple of the boundary-layer scale, $y_{\max} = m(\delta^*/L^*)$, with $m \approx 8$ –12 for the wall-trapped second mode and larger ($m \approx 35$ –45) for the more diffuse first mode; the bare values $y_{\max} = 6/12$ are only a fallback. The default resolution is $N = 128$ ($n = 129$). Because the method is spectral, the eigenvalue converges exponentially in N [4, 12]; at the default resolution the worked-example eigenvalue of Section 4.6 is already converged to many digits.

Discretising (29) is then the purely mechanical application of the substitutions (41) through the one-rule block assembly (43): each scalar entry of the 4×4 operator becomes an $n \times n$ block, and the pencil becomes the generalised problem $A\phi = cB\phi$ of Section 4.3. Equation (43) is the entire assembly: A is built from exactly three ingredients (the differentiation matrices, diagonal samples of

mean-flow profiles, and the scalars α, M_e, Re), while B , the discrete image of the algebraic operator M , marks where the eigenvalue resides. All four coefficient matrices are recorded in Section 3: M and A_2 in (30) and (31), A_1 and A_0 in (33) and (34). For compact displays this section uses the transport shorthands $m_1 = d\mu/dT$, $m_2 = d^2\mu/dT^2$, $k_1 = \frac{1}{\bar{\kappa}}d\kappa/dT$, $k_2 = \frac{1}{\bar{\kappa}}d^2\kappa/dT^2$ and the prefactors $\nu_R = 1/(Re\bar{\rho})$, $\mathcal{C}_\kappa = \gamma\bar{\mu}/(Pr Re \bar{\rho})$, $\mathcal{C}_d = \gamma(\gamma - 1)M_e^2/(Re \bar{\rho})$, $d_R = (\gamma - 1)M_e^2\nu_R$.

Figure 4 shows the resulting block occupancy; Figure 5 the assembled numbers for a small solved case. Note that α enters the operator only as $i\alpha$ and α^2 , so it is a quadratic polynomial in α ; this is exactly the property exploited in Section 4.4, where regrouping the same physics by powers of α yields the spatial quadratic pencil.

Example: the continuity block-row. Row C of the assembly rule (43) reads, term for term,

$$A_{(C,\hat{u})} = i\alpha I, \quad A_{(C,\hat{v})} = D_1 - \Lambda(\bar{T}'/\bar{T}), \quad A_{(C,\hat{T})} = -i\alpha \Lambda(\bar{U}/\bar{T}), \quad A_{(C,\hat{p})} = i\alpha\gamma M_e^2 \Lambda(\bar{U}),$$

$$B_{(C,\hat{T})} = -i\alpha \Lambda(1/\bar{T}), \quad B_{(C,\hat{p})} = i\alpha\gamma M_e^2 I$$

This gives six non-zero blocks, four in A and two in B , each a diagonal matrix or a diagonal matrix times D_1 . Every other block-row is read off (30)–(34) identically.

$A\phi = cB\phi$: each $4n \times 4n$ matrix is a 4×4 grid of $n \times n$ blocks (rows = equations, columns = fields)

A — dense operator (15 blocks)					B — the c -terms only (5 blocks)				
	$\hat{u}$	$\hat{v}$	$\hat{T}$	$\hat{p}$		$\hat{u}$	$\hat{v}$	$\hat{T}$	$\hat{p}$
continuity + state	$i\alpha I$	$D_1 - \frac{\bar{p}}{\bar{T}}$	$-i\alpha \frac{\bar{U}}{\bar{T}}$	$i\alpha\gamma M_e^2 \bar{U}$	continuity + state			$-\frac{i\alpha}{\bar{T}}$	$i\alpha\gamma M_e^2$
x-momentum	$i\alpha \bar{U}$ + visc.	$\bar{U}'$ + visc.	transport	$i\alpha \bar{p}^{-1}$	x-momentum	$i\alpha$			
y-momentum	visc.	$i\alpha \bar{U}$ + visc.	transport	$\bar{p}^{-1} D_1$	y-momentum		$i\alpha$		
energy	dissip.	$\bar{T}' + \dots$	$i\alpha \bar{U}$ $- \mathcal{C}_\kappa \mathcal{C}_c$	0	energy			$i\alpha$	

Figure 4: Block structure of $A\phi = cB\phi$. Rows: the four equations (continuity, x -momentum, y -momentum, energy); columns: the four fields $\hat{u}, \hat{v}, \hat{T}, \hat{p}$; shaded = non-zero block. A (left) is block-dense; B (right) holds only five non-zero blocks, the sparsity pattern of M in Eq. (30); every term there carries the eigenvalue c .

Real assembled operator: worked example ($M = 6$, $R = 5500$, $\alpha_L = 0.174$) — complements the symbolic block figure

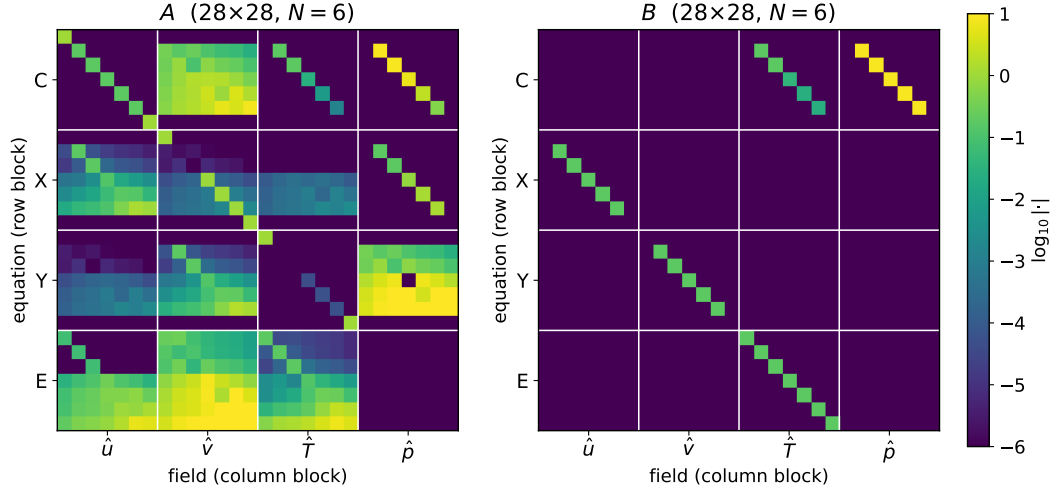


Figure 5: The same block structure assembled numerically ($N = 6$, $4n = 28$, at the parameters of Section 4.6), shown as $\log_{10}$ of the entry magnitudes. Gridlines every n rows/columns mark the C, X, Y, E row blocks and $\hat{u}$, $\hat{v}$, $\hat{T}$, $\hat{p}$ column blocks of Figure 4: A is densely populated throughout, while B carries only its five algebraic blocks.

Boundary conditions as row replacement. A collocation row states that the equation holds at node y_j ; at the wall and freestream those rows are overwritten by the boundary conditions. With the stacking $\phi = [\hat{u}; \hat{v}; \hat{T}; \hat{p}]$, field block $b \in \{0, 1, 2, 3\}$ occupies global indices $bn, \dots, bn + n - 1$, and, from the Chebyshev node ordering ($\xi_0 = +1$ at $y_{\max}$, $\xi_N = -1$ at the wall), local index 0 is the *freestream* and local index $n-1$ the *wall*:

condition	field	global row	row becomes
no-slip (wall)	$\hat{u}$	$n - 1$	identity ($\hat{u} = 0$)
no-slip (wall)	$\hat{v}$	$2n - 1$	identity ($\hat{v} = 0$)
thermal (wall)	$\hat{T}$	$3n - 1$	identity ($\hat{T} = 0$, isothermal) or $D_1[\text{wall}, :]$ ($D\hat{T} = 0$, adiabatic)
decay (freestream)	$\hat{u}$	0	identity
decay (freestream)	$\hat{v}$	n	identity
decay (freestream)	$\hat{T}$	$2n$	identity

The same rows of B are zeroed, so each constraint is purely algebraic; these zero rows are what make B singular and the problem irreducibly generalised. The pressure $\hat{p}$ carries no explicit boundary row. The inverted node ordering is a genuine implementation hazard: because $\xi_0 = +1 \rightarrow y_{\max}$ and $\xi_N = -1 \rightarrow y = 0$, index 0 is the freestream and $n-1$ the wall, the opposite of the naive guess, and exchanging them produces plausible-looking but wrong eigenvalues (Figure 6).

Boundary conditions = row replacement
in the state vector $[\hat{u}; \hat{v}; \hat{T}; \hat{p}]$

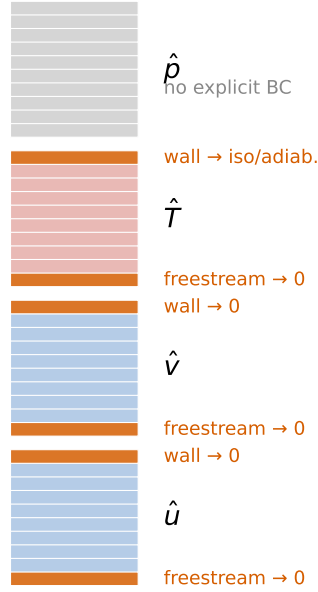


Figure 6: Boundary conditions as row replacement: the stacked state $[\hat{u}; \hat{v}; \hat{T}; \hat{p}]$ with the six overwritten rows highlighted (wall and freestream for $\hat{u}, \hat{v}, \hat{T}$); $\hat{p}$ carries no boundary row. Node index 0 is the freestream and $n-1$ the wall.

4.3 The temporal problem: a generalised eigenvalue problem

Given a real wavenumber α , the task is to find the complex phase speed c and mode shape ϕ ; the growth rate follows as $\omega_i = \alpha c_i$. With A, B from Section 4.2 this is the generalised eigenvalue problem

$$A\phi = cB\phi, \quad A, B \in \mathbb{C}^{4n \times 4n}, \quad (45)$$

solved directly by the QZ algorithm [17], which returns all $4n$ eigenvalues at once; no initial guess is required, the defining advantage of the boundary-value route [5]. The discrete eigenvector ϕ is the eigenfunction $\hat{q}(y)$ sampled at the grid points.

A disturbance $\propto e^{i\alpha(x-ct)}$ with real α behaves as $e^{\alpha c_i t}$ in time, so

$$c_i = \text{Im}(c) > 0 \Rightarrow \text{unstable (grows in time)}, \quad c_r = \text{Re}(c) = \text{phase speed}, \quad (46)$$

and the phase speed classifies the mode: a first (Tollmien-Schlichting) mode has $c_r \approx 0.3-0.6$, a second (Mack) mode $c_r \approx 1 - 1/M_e$ [1]. The raw spectrum is filtered to $\text{Re}(c) \in (-0.5, 1.5)$, $|\text{Im}(c)| < 0.5$ and sorted by c_i ; the identification of the physical mode within the filtered set is the subject of Section 4.5.

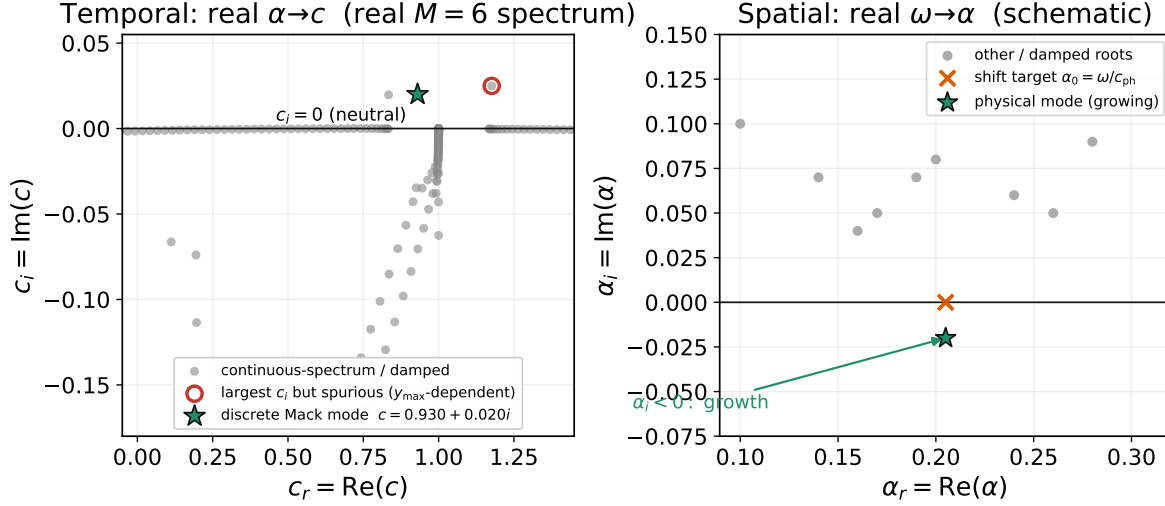


Figure 7: The two eigenvalue planes. *Left (temporal):* input real α ; the phase speeds c populate the complex plane; the physical discrete mode sits above the line $c_i = 0$ (unstable) while the continuous spectrum is damped. *Right (spatial, Section 4.4):* input real ω ; the wavenumbers α are sought near a shift target α_0 ; the physical mode has $\alpha_i < 0$, i.e. spatial growth.

4.4 The spatial problem: a quadratic eigenvalue problem

Given a real frequency ω (the physically forced case, since a fixed-frequency source generates a disturbance that grows in space), the task is to find the complex wavenumber α ; the growth rate follows as $\sigma = -\alpha_i$. Because the viscous and conduction terms carry $\partial_x^2 \mapsto -\alpha^2$, the eigenvalue now appears quadratically. Grouping the discretised equations by powers of α gives the quadratic eigenvalue problem

$$(C_0 + \alpha C_1 + \alpha^2 C_2)\phi = 0. \quad (47)$$

The spatial closure. The spatial operator is not obtained by re-grouping the temporal listing. It uses the *enthalpy* form of the energy equation, in which $\hat{p}$ appears explicitly through $D\hat{p}/Dt$, and a bulk-viscosity ratio $\lambda/\mu = 1.2$ in place of the Stokes hypothesis, giving the viscous coefficients of Mack [1, Eq. 8.5]

$$c_{\alpha^2} = \frac{2(2+\lambda/\mu)}{3} \approx 2.13, \quad c_{\alpha} = \frac{1+2\lambda/\mu}{3} \approx 1.13, \quad c_a = \frac{2(\lambda/\mu-1)}{3} \approx 0.13. \quad (48)$$

The two closures differ only in the bulk-viscosity treatment and the energy-equation arrangement; both approximate the same physical mode and agree on it to within discretisation error.

The three coefficient matrices. Because α enters the disturbance equations only as $i\alpha$ (one streamwise derivative) and α^2 (two), every term of the discretised system belongs to exactly one power of α , and sorting by that power is the entire construction of (47). The three matrices are displayed in the same 4×4 block form as the temporal operator, rows (C, X, Y, E) and columns ($\hat{u}, \hat{v}, \hat{T}, \hat{p}$); entries now contain D_1, D_2 directly, every barred coefficient stands for its diagonal sample $\Lambda(\cdot)$, and I is the identity. Besides m_1, m_2, k_1, k_2 and the prefactors $\nu_R = 1/(Re\bar{\rho})$, $d_R = (\gamma - 1)M_e^2\nu_R$, two operator shorthands keep the display compact: the viscous and conductive diffusion operators

$$\mathcal{V} = \nu_R(\bar{\mu}D_2 + \bar{\mu}'D_1), \quad \mathcal{K} = \nu_R\bar{\kappa}(D_2 + 2k_1\bar{T}'D_1 + k_1\bar{T}'' + k_2(\bar{T}')^2), \quad (49)$$

where the operator in $\mathcal{K}$ is the conduction operator $\mathcal{L}_c$ of (32) with its $-\alpha^2$ term removed: that term now lives in C_2 .

C_2 collects the $\partial_x^2 \mapsto -\alpha^2$ terms, the streamwise second derivatives of viscous diffusion and heat conduction. It is purely algebraic and has the same occupancy as A_2 , with the pressure column empty because there is no pressure diffusion:

$$C_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ c_{\alpha^2} \nu_R \bar{\mu} & 0 & 0 & 0 \\ 0 & \nu_R \bar{\mu} & 0 & 0 \\ 0 & 0 & \nu_R \bar{\kappa} & 0 \end{pmatrix}. \quad (50)$$

C_1 collects every term with a single streamwise derivative: streamwise convection at speed $\bar{U}$, the streamwise pressure gradient, and the mixed $\partial_x \partial_y$ viscous couplings. A common factor i comes out in front:

$$C_1 = i \begin{pmatrix} I & 0 & -\bar{U}/\bar{T} & \gamma M_e^2 \bar{U} \\ \bar{U} & -\nu_R(c_{\times} \bar{\mu} D_1 + \bar{\mu}') & 0 & \bar{\rho}^{-1} \\ -\nu_R(c_{\times} \bar{\mu} D_1 + c_a \bar{\mu}') & \bar{U} & -\nu_R m_1 \bar{U}' & 0 \\ 0 & -2d_R \bar{\mu} \bar{U}' & \bar{U} & -(\gamma - 1) M_e^2 \bar{U} \bar{\rho}^{-1} \end{pmatrix}. \quad (51)$$

C_0 carries everything α -free: the wall-normal physics (the D_1 , D_2 terms) and the frequency. The eigenvalue-marking role of the temporal B is now played by the $i\omega$ entries, which sit exactly where the algebraic operator $\mathbf{M}$ was non-zero, because $-i\omega$ is the ansatz image of the time derivative:

$$C_0 = \begin{pmatrix} 0 & D_1 - \bar{T}'/\bar{T} & i\omega/\bar{T} & -i\omega \gamma M_e^2 \\ -i\omega - \mathcal{V} & \bar{U}' & -\nu_R(m_1(\bar{U}'' + \bar{U}' D_1) + m_2 \bar{U}' \bar{T}') & 0 \\ 0 & -i\omega - c_{\alpha^2} \mathcal{V} & 0 & \bar{\rho}^{-1} D_1 \\ -2d_R \bar{\mu} \bar{U}' D_1 & \bar{T}' & -i\omega - \mathcal{K} - d_R m_1 (\bar{U}')^2 & i\omega(\gamma - 1) M_e^2 \bar{\rho}^{-1} \end{pmatrix}. \quad (52)$$

Equations (50)–(52) are the complete spatial operator: given ω , the mean profiles, and D_1, D_2 , the three matrices are assembled block by block exactly as in (43), the boundary rows are replaced as in Section 4.2, and (47) is solved for the complex wavenumber.

Companion linearisation. A quadratic pencil becomes an ordinary generalised problem of twice the size by introducing the auxiliary variable $\boldsymbol{\psi} = \alpha \boldsymbol{\phi}$ [18]:

$$\underbrace{\begin{bmatrix} -C_1 & -C_0 \\ I & 0 \end{bmatrix}}_{\mathcal{L}} \begin{bmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{bmatrix} = \alpha \underbrace{\begin{bmatrix} C_2 & 0 \\ 0 & I \end{bmatrix}}_{\mathcal{R}} \begin{bmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{bmatrix}. \quad (53)$$

The lower block row enforces $\boldsymbol{\psi} = \alpha \boldsymbol{\phi}$; the upper row then reproduces (47). The eigenvalue is α and the physical shape is the lower half $\boldsymbol{\phi}$: since $\boldsymbol{\psi} = \alpha \boldsymbol{\phi}$ the two halves are parallel, and by convention the lower half is normalised as the eigenfunction (Figure 8).

Companion linearisation: a quadratic EVP becomes an ordinary EVP of double size

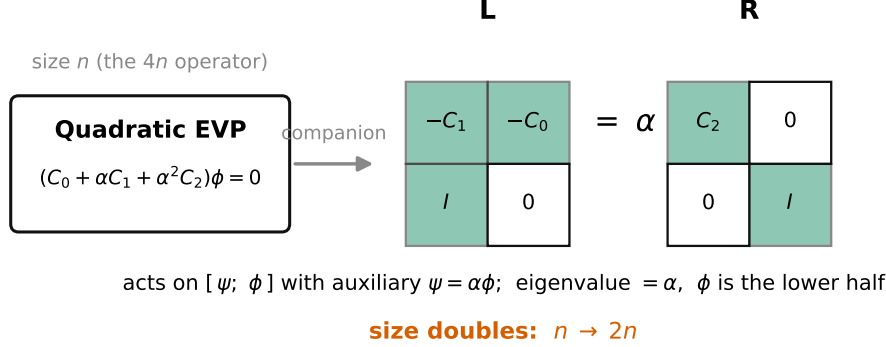


Figure 8: Companion linearisation (53): the $4n$ -sized quadratic problem becomes an ordinary generalised problem of size $8n$ in $[\psi; \phi]$ with $\psi = \alpha\phi$. Its eigenvalue is exactly α ; the eigenfunction is the lower half.

Shift–invert. The physically relevant branch is targeted with a shift $\alpha_0 = \omega/c_{\text{ph}}$ ($c_{\text{ph}} = 1 - 1/M_e$ for $M_e > 3$, else ≈ 0.4 ; these are the second- and first-mode phase-speed estimates of Section 4.3). Subtracting α_0 and inverting maps eigenvalues near α_0 to large μ :

$$\text{eigenvalues of } (\mathcal{L} - \alpha_0 \mathcal{R})^{-1} \mathcal{R} \text{ are } \mu = \frac{1}{\alpha - \alpha_0}, \quad \Rightarrow \quad \alpha = \alpha_0 + \frac{1}{\mu}, \quad (54)$$

so the largest- $|\mu|$ modes are the wavenumbers nearest the physical one (in our LST solver, a dense solve followed by selecting the 20 eigenvalues nearest α_0 and sorting by $\text{Im } \alpha$). The amplitude varies as $e^{i\alpha x} = e^{i\alpha_r x} e^{-\alpha_i x}$, so

$$\sigma = -\text{Im}(\alpha) = -\alpha_i > 0 \Rightarrow \text{unstable (grows downstream)}. \quad (55)$$

Alternative route: Newton iteration from a temporal seed. When shift–invert struggles, a more robust path solves the *temporal* problem at a real trial wavenumber $\alpha_r = \omega/c_{\text{est}}$ ($c_{\text{est}} = 0.98$ for $M_e > 3$, else 0.35), keeping only modes that agree across two resolutions (tolerance 8×10^{-3}); forms the seed $\text{Im}(\alpha) \approx -\alpha_r c_i / c_r$, a phase-speed estimate of the rigorous group-velocity conversion of Gaster [16] that suffices here because it only seeds the iteration, and then Newton-iterates the dispersion residual $F(\alpha) = \alpha c(\alpha) - \omega = 0$, $\alpha^{(k+1)} = \alpha^{(k)} - F/F'$, with F' approximated by a finite difference (step 10^{-5} , tolerance 10^{-6} , at most 20 iterations). Both routes return the same $\sigma = -\alpha_i$.

4.5 Identifying the physical mode

A single solve of (45) returns $4n$ eigenvalues, several hundred at the default resolution, of which typically *one* is the boundary-layer mode sought. The remainder fall into three recognisable populations:

- **Infinite eigenvalues.** The boundary-condition rows carry no eigenvalue and leave zero rows in B , so QZ reports the corresponding pairs as infinite or absurdly large; the raw band filter of Section 4.3 removes them.
- **The discretised continuous spectrum.** On the unbounded domain the freestream solutions form continuous branches rather than isolated points; the finite grid renders those branches as dense strings of grid-dependent eigenvalues.

- **Discrete modes.** A handful of isolated eigenvalues whose eigenfunctions are trapped inside the layer; the unstable ones among them are the physics sought.

No property of the eigenvalue alone separates the last two populations: in the worked example of Section 4.6 the largest growth rate in the entire filtered spectrum belongs to a continuous-spectrum artifact. The separation of the discrete modes from the continuous spectrum rests on the *eigenfunction* and the grid [19, 20], through two implemented tests.

Test 1: freestream decay. A genuine boundary-layer eigenfunction decays exponentially above the layer, whereas a continuous-spectrum eigenfunction oscillates undiminished to the top of the box [20]. The implemented measure is the edge ratio of the stacked eigenvector,

$$r_e = \frac{\langle a(y_j) \rangle_{4 \text{ outermost nodes}}}{\max_j a(y_j)}, \quad a(y_j) = \max(|\hat{u}_j|, |\hat{v}_j|, |\hat{T}_j|, |\hat{p}_j|), \quad (56)$$

and candidates with $r_e \geq 0.1$ are rejected. The test is not marginal. At the worked-example parameters ($M_e = 6$, $R = 5500$, $\alpha_L = 0.174$) the accepted Mack mode has $r_e = 1.7 \times 10^{-7}$, while the two rejected amplified candidates have $r_e = 0.34$ and 0.50 : six orders of magnitude of separation.

Test 2: domain stationarity. A genuine eigenvalue is a property of the boundary layer, not of the box that truncates it, so raising $y_{\max}$ must leave it fixed; the continuous-spectrum approximants, which owe their existence to the unbounded domain [19], rearrange with the grid. The solve is repeated on a taller domain and a candidate is kept only if its c reappears there within 2×10^{-3} . At the worked-example parameters, raising $y_{\max}$ from 10 to 12 (δ^*/L^*) moves the Mack mode by 5×10^{-8} and every rejected candidate by 10^{-3} – 10^{-2} .

4.6 Worked example: one temporal solve

Consider a strong Mack-mode case: $M_e = 6.0$, $R = 5500$, $\alpha_L = 0.174$, adiabatic wall, L^* scaling, $N = 180$ ($n = 181$, matrices $4n = 724$ wide), and domain height $y_{\max} = 10(\delta^*/L^*) \approx 128$ (a multiple of the boundary-layer scale, per Section 4.2). Solving (45) returns all 724 eigenvalues at once, and the two acceptance tests of Section 4.5, freestream decay of the eigenfunction and stationarity of c under $y_{\max} \rightarrow 12(\delta^*/L^*)$, sift the one physical mode from the crowd.

The largest-growth eigenvalue is a decoy: it sits at $c_r = 1.18$, keeps a third of its peak amplitude at the top of the box, drifts by 6×10^{-3} when the domain is raised, and is unphysically supersonic ($c_r > 1$); it belongs to the continuous spectrum, not the discrete set. The physical answer is the Mack (second) mode $c = 0.930 + 0.0200i$, whose eigenfunction decays cleanly at the freestream (Figure 9) and whose phase speed is stationary to one part in 10^7 when the box is enlarged. Its growth rate $\omega_i = \alpha c_i = +0.00348$ is positive, so the wave amplifies, and its phase speed $c_r = 0.93$ is the second-mode value near $1 - 1/M_e$ [1]. This single $\omega_i > 0$ is one point inside the neutral curve; repeating the solve across an (R, α) grid and locating the zero crossing of ω_i traces the neutral curve of Section 4.7.

Computed eigenfunction: $M = 6$ Mack mode, $c = 0.930 + 0.020i$

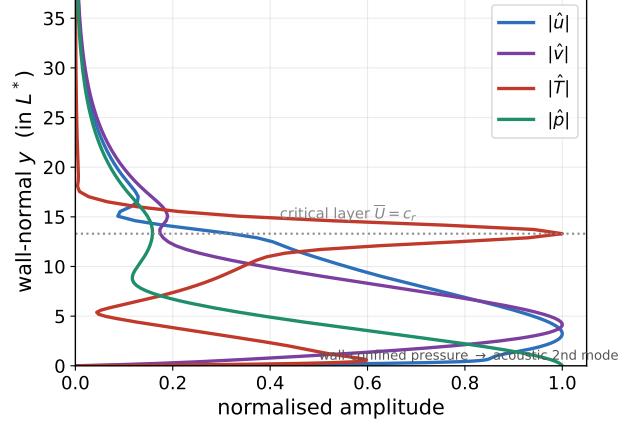


Figure 9: Computed eigenfunction of the worked-example Mack mode: the four amplitude fields $|\hat{u}|$, $|\hat{v}|$, $|\hat{T}|$, $|\hat{p}|$ against y (each normalised). $|\hat{T}|$ peaks at the critical layer ($\bar{U} = c_r$); $|\hat{p}|$ is wall-confined, the acoustic signature of the second mode. All fields decay at the freestream, confirming a discrete mode.

4.7 Neutral curves

The neutral curve is the boundary between growth and decay: the locus $c_i(R, \alpha) = 0$ (temporal) or $\sigma(R, \omega) = 0$ (spatial), inside which the flow is unstable. Our LST solver traces it in three steps:

1. **Sweep** an (R, α) grid at fixed M_e ; at each node solve (45), apply the mode tests of Section 4.5, and store c_i .
2. **Extract** the zero contour of the stored field by marching squares; unresolved nodes are tagged stable so that the contour closes.
3. **Continuation** for difficult branches: at low Mach number the first-mode eigenvalue approaches the continuous spectrum and the grid sweep loses it. A continuation pass then tracks the mode by proximity: from a clean seed it marches in R , follows the nearest eigenvalue, and secant-solves $c_i = 0$ at each step, each converged eigenvalue seeding the next parameter value.

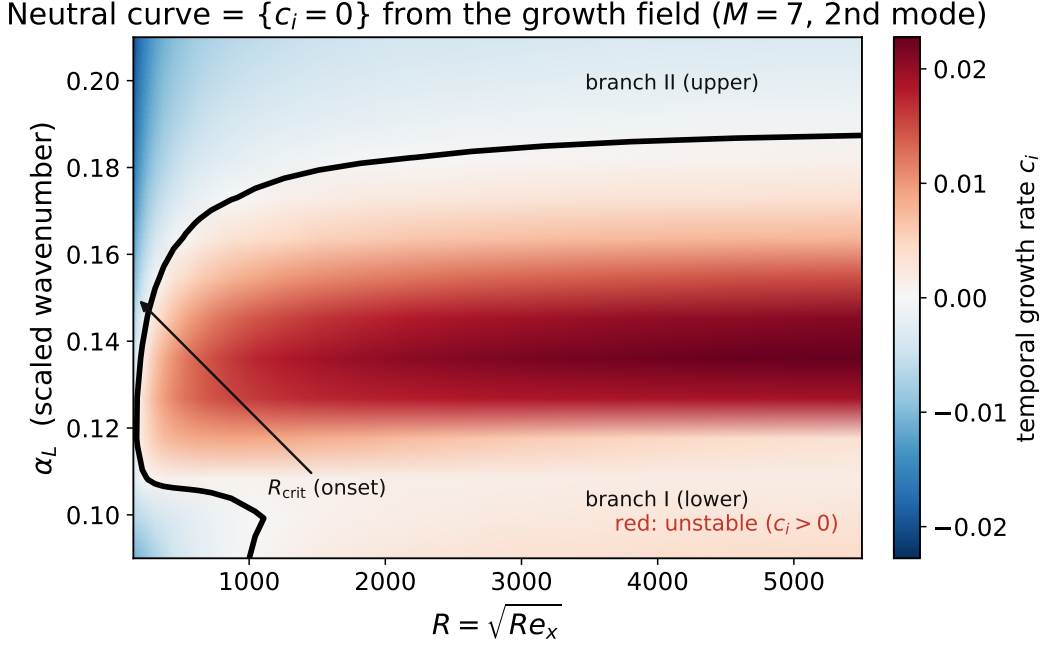


Figure 10: Steps 1–2 of the neutral-curve procedure for a solved case ($M_e = 7$, second mode): the coloured field is the growth rate $c_i(R, \alpha)$ from the sweep; the black curve is the extracted $\{c_i = 0\}$ contour. The flow is unstable inside the loop; R_{crit} marks the lower branch (branch I) where instability first appears as R increases, and the curve closes at branch II.

4.8 N -factor and transition

What matters for transition is the *accumulated* amplification of a fixed-frequency disturbance. For a chosen reduced frequency F , the spatial problem of Section 4.4 is solved at successive stations to obtain $\sigma_L = -\text{Im}(\alpha_L)$, the spatial growth rate (55) expressed in the L^* scaling, and integrated from the lower neutral point R_0 where σ_L first becomes positive:

$$N(R) = \int_{R_0}^R \max(\sigma_L, 0) dR' \quad (\text{trapezoidal}), \quad \frac{A(R)}{A(R_0)} = e^{N(R)}. \quad (57)$$

The amplitude grows by e^N , and transition is empirically expected near $N \approx 9$ –11, following the e^N method of Smith & Gamberoni and van Ingen [2, 3]. For a cone, surface arc length is mapped to an equivalent R before integration. The transition N is a correlation, not a first-principles criterion: it depends on the disturbance environment (flight ~ 9 –11; noisy conventional tunnels transition at $N \sim 4$ –6), presumes a single dominant linear mode, and absorbs the unknown initial amplitude set by receptivity.

5 Validation and verification

5.1 Summary of all cases

Table 1 lists every case presented below, together with the solver route that produced the compared quantity: the formulation (temporal or spatial), the numerical route (collocation eigenvalue solve or shooting), and its input and output. The routes themselves are specified in Section 4; the caveats live in each subsection. The Malik temporal test cases 3, 4, and 5, previously logged non-reproducible for want of their dimensional conditions, are now included, once Malik’s own Table I supplied the missing T_0 and wall temperatures.

Table 1: Cases presented in this document, with the solver route behind each comparison. GEVP: the temporal generalised eigenvalue problem $A\phi = cB\phi$ (input a real wavenumber, output the complex spectrum); QEP: the spatial quadratic eigenvalue problem in α (input a real frequency, output the complex wavenumber); shooting: the exact first-order march with $\sigma_{\min} \rightarrow 0$ root search, an independent route documented separately.

Case	Source	Quantity	Solver route (input $\rightarrow$ output)
<i>Incompressible core</i>			
Plane Poiseuille	Orszag [4]	O–S eigenvalue c	temporal O–S GEVP, QZ ($\alpha, Re \rightarrow$ least-stable c)
OS spectrum (32 modes)	Orszag [4], Tab. 5	full temporal spectrum	temporal O–S GEVP, QZ ($\alpha, Re \rightarrow$ full c spectrum)
<i>Compressible eigenvalues</i>			
Malik case 6	Malik [5], Tab. IX	spatial α (2nd mode)	spatial QEP, shift–invert ($\omega \rightarrow \alpha$)
Malik cases 3/4/5	Malik [5], Tab. IV/V/VI	temporal ω (M2.5/10)	temporal GEVP, 3-D for the oblique case 3 ($\alpha[, \beta] \rightarrow \omega$)
Malik Table X	Malik [5], Tab. X	spatial α (M10 2nd mode)	spatial QEP, shift–invert ($\omega \rightarrow \alpha$)
Malik Fig. 4 eigenfn.	Malik [5], Fig. 4	M10 2nd-mode $\hat{T}(y)$ shape	temporal GEVP, eigenvector ($\alpha \rightarrow \hat{q}(y)$)
B&M branches	Balakumar–Malik [20]	discrete/continuous topology	spatial QEP, full companion spectrum ($\omega \rightarrow$ all α)
B&M via Xi–Ren–Fu	Balakumar–Malik [20]	spatial α (2nd mode)	spatial QEP ($\omega \rightarrow \alpha$)
<i>Second mode: curves & families</i>			
Mack Fig. 10.6	Mack [1]	$\omega_{i,\max}(R)$, M4.5–10	temporal GEVP swept in α per R ($\alpha \rightarrow \omega_i$; max over α)
Egorov M6	Egorov et al. [22]	most-amplified ω	spatial QEP swept in ω ($\omega \rightarrow \sigma = -\alpha_i$)
Ma & Zhong M4.5	Ma & Zhong [21]	neutral curve, spatial 2nd mode	spatial QEP at fixed F ($\omega \rightarrow \alpha_i$; $\sigma = 0$ crossings)
<i>First mode: curves & families</i>			
Mack Fig. 10.1	Mack [1]	neutral loop, M1.6	temporal GEVP on an (R, α) grid ($\alpha \rightarrow c_i$; $c_i = 0$ contour)
Mack Fig. 10.3	Mack [1]	$\omega_{i,\max}(R)$, M1.3/3.0	exact oblique shooting, continuation from an anchor root ($\alpha, \beta \rightarrow c$ via $\sigma_{\min} \rightarrow 0$)
Mack Fig. 10.4	Mack [1]	oblique 1st mode, M4.5–10	temporal 3-D GEVP ($\alpha, \psi \rightarrow \omega_i$; max over α, ψ)
<i>Mixed first + second mode</i>			
Özgen Fig. 3	Özgen–Kırcah [6]	neutral-curve family (both modes), M2–10	temperature-form temporal GEVP + branch continuation ($\alpha_L \rightarrow c_i$; $c_i = 0$ contour)

5.2 Incompressible core

5.2.1 Orszag (1971): plane-Poiseuille Orr–Sommerfeld eigenvalue

Reference. Orszag [4], the canonical plane-Poiseuille temporal eigenvalue.

Conditions. Plane Poiseuille flow $U = 1 - y^2$ on $[-1, 1]$, Reynolds number $Re = 10\,000$, stream-wise wavenumber $\alpha = 1$, temporal formulation.

Solving the temporal Orr–Sommerfeld problem by Chebyshev collocation and dense QZ (input the real wavenumber α and Re , output the complex phase-speed spectrum), Our LST code reproduces the benchmark least-stable eigenvalue (phase speed) $c = 0.23753 + 0.00374i$ to an absolute error below 10^{-4} at $N = 64$, tightening toward $\sim 10^{-7}$ under refinement (Table 2); this is a scalar comparison against an essentially exact reference, so no overlay figure is needed. The same solver also recovers the Blasius Tollmien–Schlichting mode at $Re_{\delta^*} = 998$, $\alpha = 0.179$ ($c_r \approx 0.36$, unstable) and finds unstable growth at $Re = 1000$, confirming the mapped semi-infinite domain and the growth-rate optimisation loop.

Table 2: Plane Poiseuille temporal eigenvalue, $Re = 10\,000$, $\alpha = 1$. Target $c = 0.23753 + 0.00374i$ [4].

N	Our LST code $c = c_r + ic_i$	$ c - c_{\text{target}} $
64	$0.23753 + 0.00374i$	$\lesssim 10^{-4}$
96	$0.23753 + 0.00374i$	$\lesssim 10^{-5}$
128	$0.23753 + 0.00374i$	$\lesssim 10^{-7}$

5.2.2 Orszag (1971) Table 5: the full Orr–Sommerfeld spectrum

Reference. Orszag [4], Table 5: the 32 least-stable Orr–Sommerfeld eigenvalues of plane Poiseuille flow at $\alpha = 1$, $R = 10\,000$, each classified symmetric (S) or antisymmetric (A).

The same temporal Orr–Sommerfeld route as Section 5.2.1 is exercised on its full-spectrum output, returning *all* eigenvalues of the collocated pencil at once rather than just the single least-stable mode. Our LST code’s dense eigensolver reproduces *all 32* tabulated eigenvalues to a maximum absolute error 7×10^{-6} (median 5×10^{-8}) at $N = 128$. It resolves the A-branch ($c_r \sim 0.96$ downward), the P-branch ($c_r \sim 0.2$ – 0.5), and the vertical S-branch, including the near-degenerate S/A pairs that differ only in the fifth–sixth digit (e.g. modes 2 and 3 at 0.96463 vs. 0.96464), exercising the dense full-spectrum path (Figure 11).

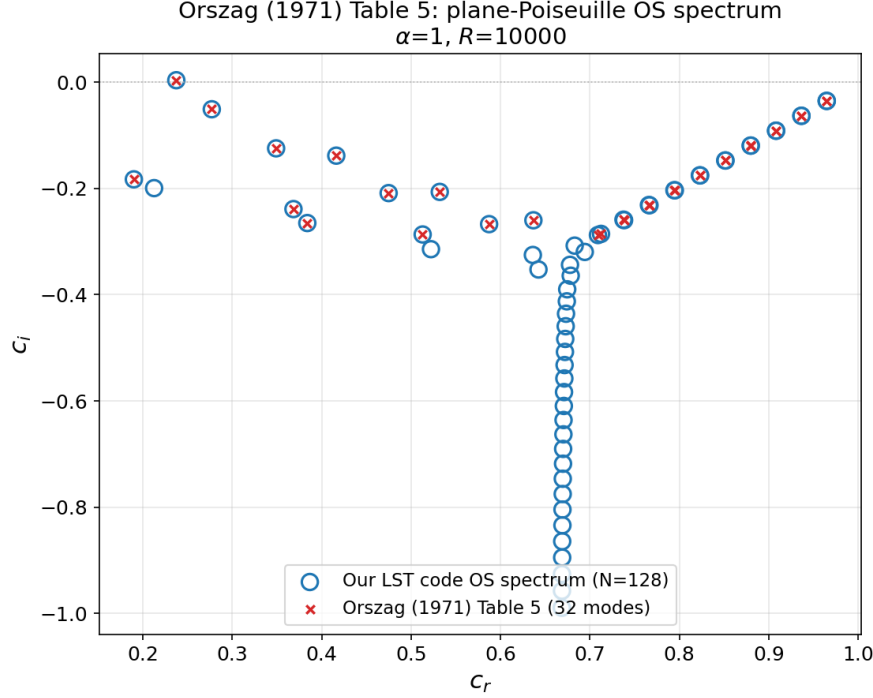


Figure 11: Orszag Table 5 [4]: Our LST code’s plane-Poiseuille Orr–Sommerfeld spectrum (circles) over the 32 tabulated eigenvalues (crosses), $\alpha = 1$, $R = 10000$. The classic “Y”-shaped spectrum (A-, P-, and S-branches), with every tabulated mode matched to $< 10^{-5}$.

5.3 Compressible eigenvalues

5.3.1 Malik (1990): compressible eigenvalues

Reference. Malik [5], the standard tabulated compressible-LST test-case suite. Case 6 (Table IX, the $M = 4.5$ spatial second mode) is the primary anchor, cross-checked digit-for-digit against Xi, Ren & Fu [23] and Hildebrand et al. [24]. Cases 3, 4, 5 (Tables IV/V/VI) and the grid-refinement case of Table X were long non-reproducible for want of their dimensional conditions ([24] lists only M and R), until Malik’s own Table I supplied the missing T_0 and wall temperatures, in particular case 4’s *cooled* wall $T_w/T_{adb} = 0.1$ that earlier attempts had assumed adiabatic.

Conditions. All cases share $Pr = 0.7$, $\gamma = 1.4$, Sutherland $S = 110.33$ K, L^* length scale. The suite spans $M = 2.5$ –10, temporal (fixed real α , complex ω) and spatial (fixed real ω , complex α) formulations, and adiabatic and cooled walls. Case 3 ($M = 2.5$, oblique $\beta = 0.1$) is a first mode; the rest are second (Mack) modes. Each anchor is Chebyshev-collocated at $N = 120$ –280 points (larger N for the thick $M = 10$ layers), domain height $y_{\max} = 40$ –140 δ^*/L^* , L^* scaling, with an isothermal disturbance-temperature condition and bulk-viscosity ratio $\lambda/\mu = 1.2$.

Both formulations of the collocation route appear, matched case by case to what Malik tabulates: the temporal cases 3–5 use the temporal GEVP (its 3-D variant for the oblique case 3), returning the complex frequency ω from a real wavenumber, while the spatial cases 6 and Table X use the spatial quadratic eigenvalue problem in companion form with shift–invert targeting, returning the complex wavenumber α from a real frequency. Table 3 lists the five anchors. Every real part (α_r or ω_r) is essentially exact, fixing each mode and its phase speed. The one outlier is Table X’s

spatial α_i , which reflects the documented hypersonic-second-mode inter-method spread: Tumin’s recomputation of case 6’s α_i from Malik’s printed value differs in the same way, and the solver of [23] gives -0.002780 . The *temporal* $M = 10$ growth rates (cases 4, 5) match closely, so the spatial α_i is the more formulation-sensitive quantity. Case 5 is Malik’s deliberately near-neutral “severe test”: his own SDSP scheme flips ω_i negative at coarse resolution, yet Our LST code lands on his converged value (Figure 12).

Table 3: The Malik [5] compressible eigenvalues. Spatial cases (6, X) compare the complex wavenumber α ; temporal cases (3–5) compare the complex frequency ω .

Case	Conditions	Our LST code	Malik
6 (Tab. IX)	$M4.5$ adiab., spatial 2nd	$0.253400-0.002490 i$	$0.253405-0.002492 i$
3 (Tab. IV)	$M2.5$ oblique, temporal 1st	$0.036735+0.000584 i$	$0.036734+0.000584 i$
4 (Tab. V)	$M10$ cooled, temporal 2nd	$0.097457+0.001988 i$	$0.097484+0.002030 i$
5 (Tab. VI)	$M10$ adiab., temporal 2nd	$0.115862+0.000156 i$	$0.115865+0.000153 i$
X (Tab. X)	$M10$ cooled, spatial 2nd	$0.095803-0.002403 i$	$0.095933-0.002156 i$

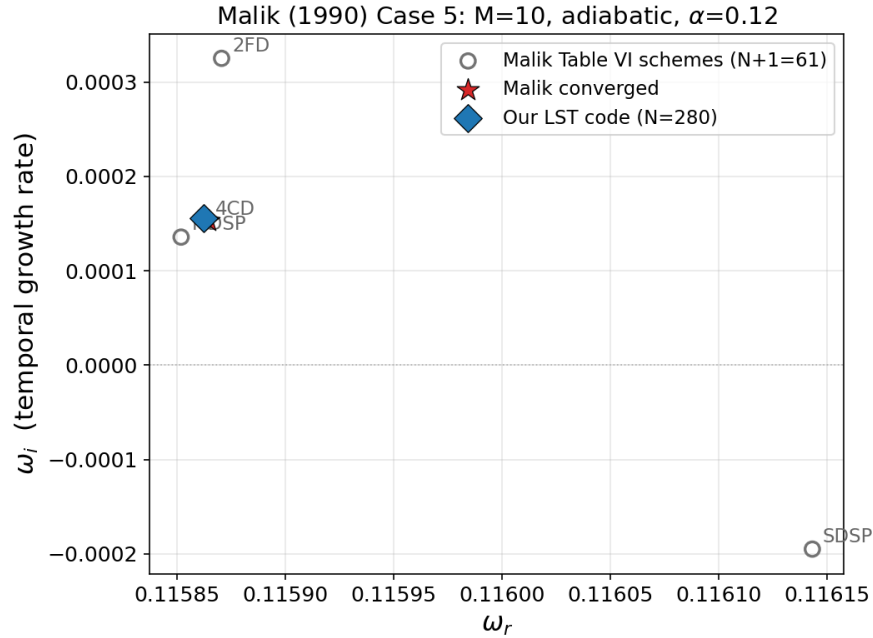


Figure 12: Malik case 5 [5] ($M = 10$ adiabatic, the deliberately near-neutral “severe test”): Our LST code (diamond) lands on Malik’s converged value (star) and his 4CD/MDSP cluster, while his own 2FD scheme sits high and SDSP flips ω_i negative; Our LST code reproduces the case Malik built to break solvers.

5.3.2 Malik (1990) Fig. 4: M10 second-mode eigenfunction

Reference. Malik [5], Fig. 4: the disturbance eigenfunctions of the same $M = 10$ cooled-wall condition as case 4. The comparison uses the temperature eigenfunction $\hat{T}$ (real and imaginary parts), the dominant, cleanly traceable curves; the small velocity and pressure eigenfunctions overlap near

the axis and are not reliably digitisable, so they are omitted.

At the case-4 conditions ($N = 200$ Chebyshev points, $y_{\max} = 75 \delta^*/L^*$, L^* scaling, isothermal disturbance-temperature condition) the temporal GEVP is read for its *eigenvector* rather than its eigenvalue: from a real wavenumber α it returns the discrete eigenpair, of which the temperature component $\hat{T}(y)$ is compared. This is therefore a mode-shape check, not an eigenvalue one, and the complex temperature eigenfunction $\hat{T}(y)$ is normalised so that $\hat{T}_r$ peaks at +1 (Malik's convention, which fixes the eigenvector's single global phase). Both the real and imaginary temperature eigenfunctions lie on the digitised Fig. 4 points (Figure 13). Our LST code's $\hat{T}_r$ traces the near-wall negative lobe, the zero crossing near $y \approx 6$, and the sharp unit peak at $y = 13.3$ ($\approx \delta^*$) against Malik's ≈ 13 ; $\hat{T}_i$ reproduces the same double-lobed shape, a positive hump near $y \approx 11$ and a negative dip near $y \approx 15$. Under the *single* global phase that makes $\hat{T}_r$ real at its peak, *both* parts fall on the reference, so the temperature eigenvector is validated in full, not merely its peak location. The temperature fluctuation dominates the velocity ($|\hat{T}|/|\hat{u}| \approx 14$), the hallmark of the hypersonic Mack mode.

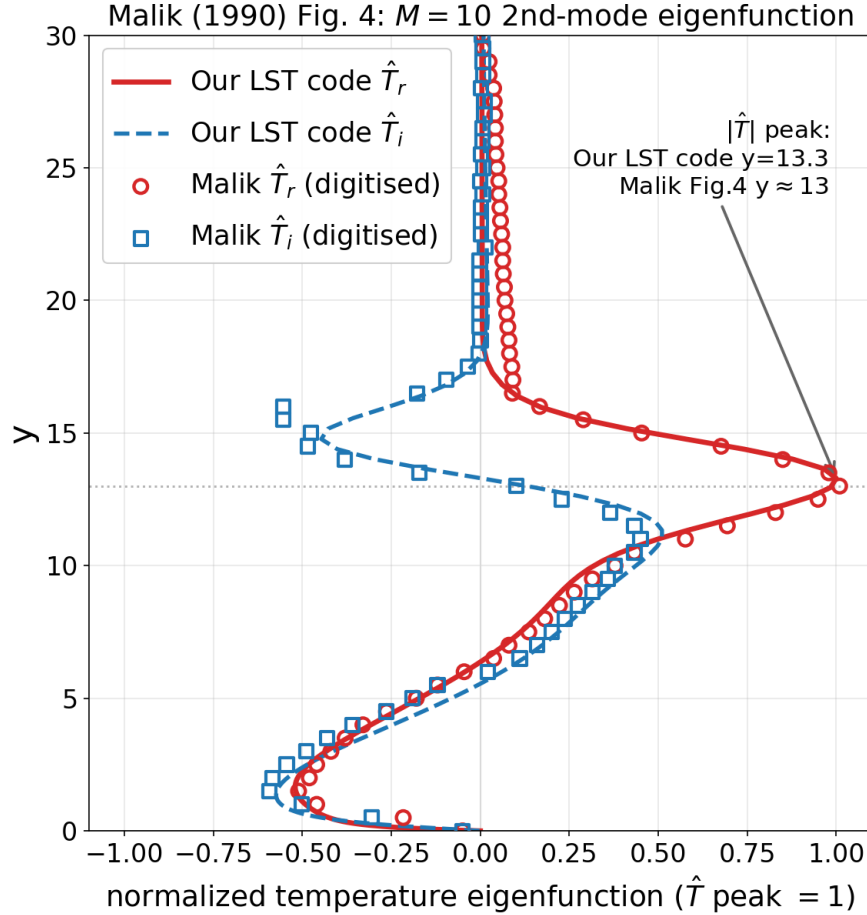


Figure 13: Malik Fig. 4 [5], $M = 10$ second-mode temperature eigenfunction: Our LST code's $\hat{T}_r$ (solid) and $\hat{T}_i$ (dashed) curves over the digitised points, normalised so $\hat{T}_r$ peaks at +1. Both parts match under a single global phase: $\hat{T}_r$ has its unit peak at $y = 13.3$ against Malik's ≈ 13 , and $\hat{T}_i$ shows the positive hump near $y \approx 11$ and the negative dip near $y \approx 15$.

5.3.3 Balakumar & Malik (1992): discrete/continuous spectrum topology

Reference. Balakumar & Malik [20]. The paper presents its branches graphically; no tabulated branch eigenvalues are openly recoverable, so this is a qualitative branch-topology check plus a fully specified spatial second-mode cross-check.

Conditions. $M = 4.5$, $T_0 = 311$ K, $Re = 1000$, $\omega = 0.2$, adiabatic (insulated) wall. The full companion spectrum is computed by Chebyshev collocation at $N = 120$ (121 nodes), $y_{\max} = 40 \delta^*/L^*$, L^* scaling, with an isothermal disturbance-temperature condition and $\lambda/\mu = 1.2$.

The spatial quadratic eigenvalue problem is solved for its *entire* companion spectrum, not merely the shift-invert neighbourhood: from a real frequency ω it returns all complex wavenumbers α of the discretised operator at once, discrete mode and rendered continuous spectrum together, which is what a branch-topology comparison requires. The comparison is thus twofold: the branch topology (how the discrete second mode separates from the continuous spectrum in the complex- α plane) and the discrete spatial eigenvalue α . Our LST code's full spatial spectrum reproduces the published topology: the discrete second mode sits at $\alpha = 0.220210 - 0.002787i$ ($c = 0.9082$), matching the published $0.220 - 0.003091i$, and is the unique least-damped root in the second-mode phase-speed band, while a cluster of roots accumulates at the freestream phase speed $c \rightarrow 1$ (the continuous spectrum). The discrete mode is set apart from that cluster in phase speed and growth rate, not by a wide gap in the complex plane: the discretised continuum is dense in α_r , so the nearest continuum root lies close by, the expected signature of the continuous spectrum crowding the discrete mode. Treated purely as a spatial second-mode anchor, the real part is essentially exact and the imaginary part carries the same inter-method α_i spread seen on Malik case 6 (Figure 14).

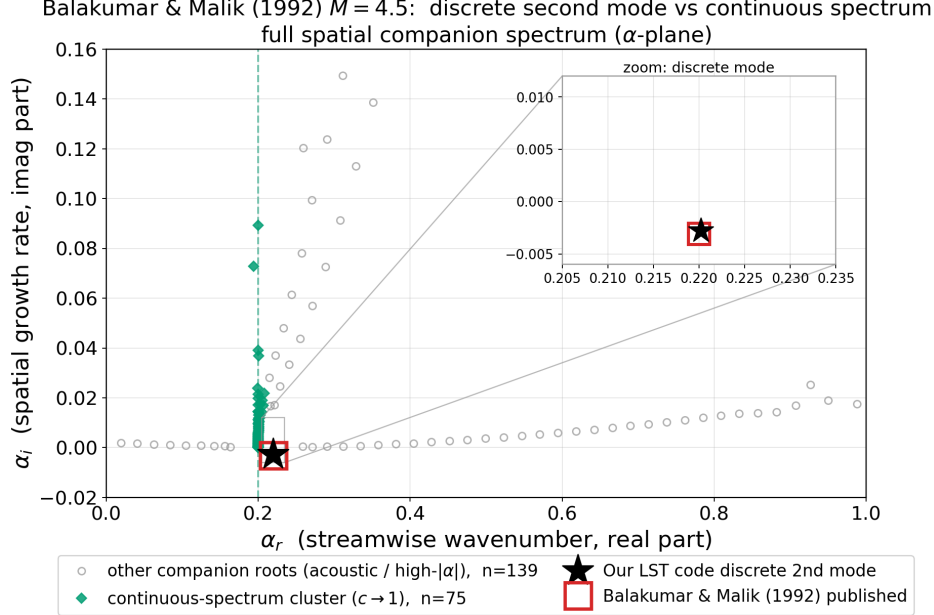


Figure 14: Balakumar & Malik [20], $M = 4.5$: Our LST code's complex- α spectrum showing the discrete second mode set apart (in phase speed and growth rate) from the continuous-spectrum cluster near $c \rightarrow 1$.

5.4 Second mode: curves and families

Having pinned the compressible operator at single points, the following cases test full growth-rate and neutral curves; the second mode is Our LST code’s design target.

5.4.1 Mack (1984) Fig. 10.6: second-mode growth vs. Reynolds number

Reference. Mack [1], Fig. 10.6.

Conditions. Air, adiabatic wall, 2-D wave ($\psi = 0^\circ$), at $M = 4.5, 5.8, 7$, and 10 . Two modelling choices make the comparison defensible: the cold edge state of Mack’s Table 11.1 [1] (resolving a historical $\sim 6\times$ gap that was an edge-temperature error) and a wall-normal domain $y_{\max} \approx 4\delta^*/L^*$ that grows with Mach ($y_{\max} = 40$ at M4.5 up to 140 at M10; a fixed short box starves the thick high-Mach boundary layer). The eigenvalue solves use Chebyshev collocation at $N = 110$, L^* scaling, and the 2-D Stokes closure $\lambda/\mu = 0$.

Sweeping the temporal GEVP in wavenumber at each Reynolds number (a real α scan at fixed R , from which the second-mode $\omega_i = \alpha c_i$ is maximised over α , then repeated over R) traces the curve. Across $M = 4.5, 5.8, 7$, and 10 , Our LST code’s maximum second-mode temporal growth rate $\omega_{i,\max}(R)$ (Mack’s L^* scale, $c_r \approx 0.9$) tracks the digitised Fig. 10.6 points. The match is tightest at $M = 7$ and $M = 10$ and carries a small residual at $M = 4.5$ and $M = 5.8$; because both ends of the Mach range match well, that residual is local rather than a monotonic high-Mach under-prediction. The $M = 4.5$ and $M = 10$ curves are shown as representatives (Figures 15 and 16); $M = 5.8$ and $M = 7$ follow the same trend and are summarised in Table 1.

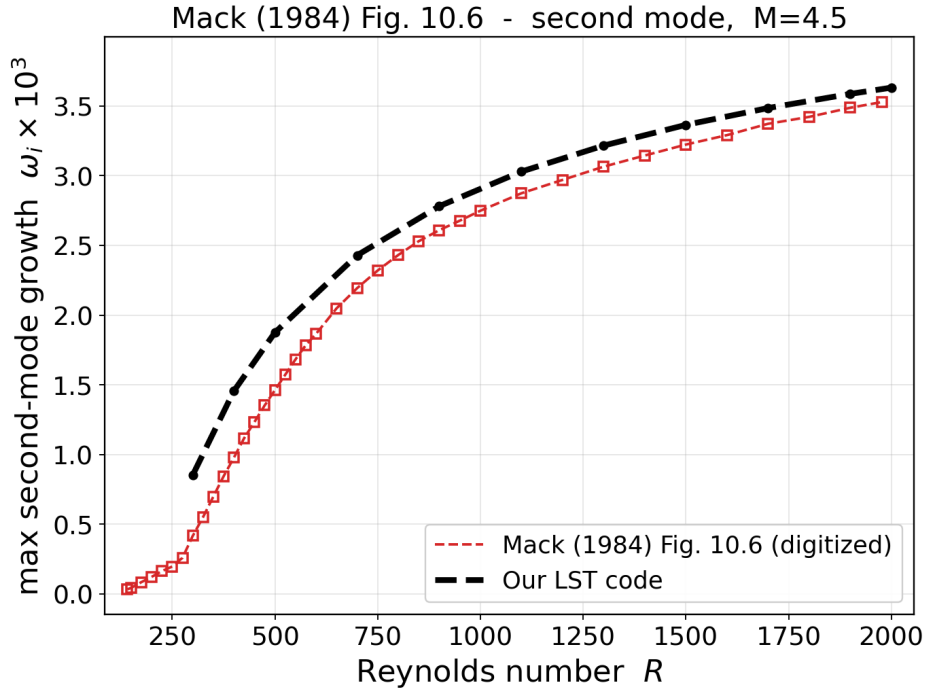


Figure 15: Mack Fig. 10.6 [1], $M = 4.5$: Our LST code’s $\omega_{i,\max}(R)$ over the digitised points, with a small residual at this Mach number.

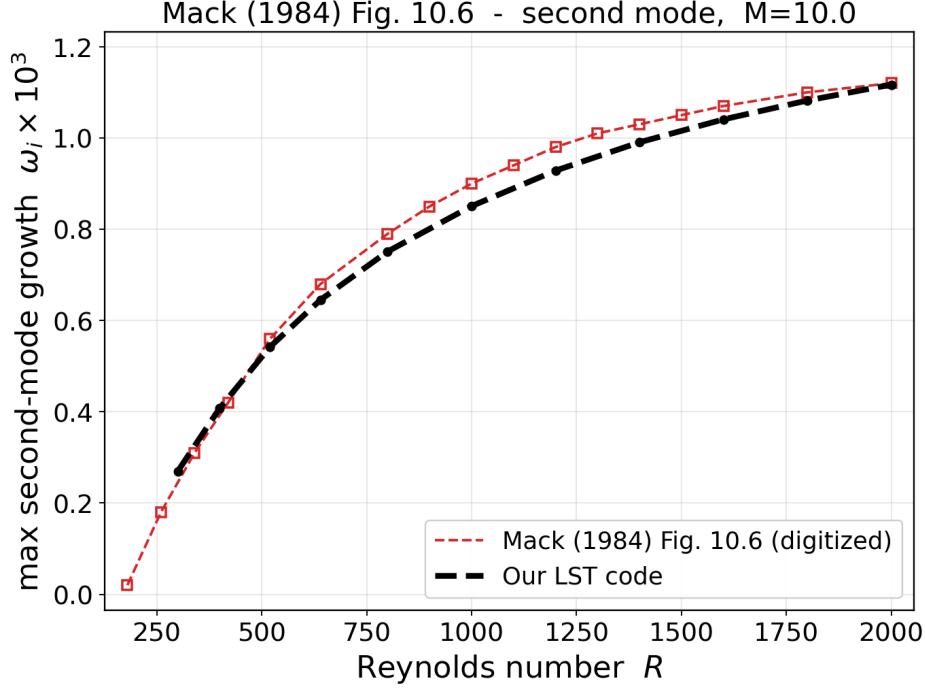


Figure 16: Mack Fig. 10.6 [1], $M = 10$: the same comparison at the top of the Mach range, with the domain-converged ($y_{\max} = 140$), cold-edge second mode.

5.4.2 Egorov et al. (2006): Mach-6 flat-plate second mode

Reference. Egorov, Fedorov & Soudakov [22]: 2-D DNS second-mode growth over a Mach-6 adiabatic flat plate, reported to agree with LST.

Conditions. Perfect gas $\gamma = 1.4$, $Pr = 0.72$, power-law viscosity $\mu/\mu_e = (T/T_e)^{0.7}$; $M_\infty = 6$, $Re_L = 2 \times 10^6$, adiabatic wall; mode-S circular frequency $\omega \approx 200$ (plate-length units) at the synchronisation location $x = 1$ ($R = \sqrt{2} \times 10^6 = 1414$). The spatial solve uses Chebyshev collocation at $N = 140$, $y_{\max} = 40 \delta^*/L^*$, L^* scaling, an isothermal disturbance-temperature condition, and $\lambda/\mu = 1.2$.

The spatial QEP is swept in frequency at the synchronisation location $R = 1414$ (real ω across the candidate band, returning the complex wavenumber α and hence the spatial growth $\sigma = -\alpha_i$ and its maximiser). Because the paper presents the DNS-vs-LST comparison as a growth-rate-vs- x curve with no printed growth-rate digit, the comparison is made on the second-mode instability band and the frequency of the most-amplified mode. Our LST code's discrete second mode ($c_r \approx 0.92$ – 0.94) is unstable across $\omega_E \in [140, 260]$, which brackets the reference forcing $\omega = 200$, and is most-amplified at $\omega_E \approx 215$; at exactly $\omega = 200$ the spatial growth is $-\alpha_i = 2.90 \times 10^{-3}$ with $c_r = 0.934$. Our LST code thus places the M6 second-mode instability squarely in the stated band, with the most-amplified frequency essentially on the reference forcing choice.

5.4.3 Ma & Zhong (2003) Fig. 15: M4.5 second-mode neutral curve

Reference. Ma & Zhong [21], §6 / Fig. 15: full neutral curves $\omega(R)$ for the first and second mode at $M = 4.5$, with the second-mode branch points quoted in text at $R = 806$ (branch I) and $R = 999.6$ (branch II) for $F = 2.2 \times 10^{-4}$.

Conditions. Air, adiabatic *mean* wall, 2-D ($\psi = 0^\circ$), $M = 4.5$, $T_e = 65.15$ K, unit $Re = 7.2 \times 10^6/\text{m}$, Sutherland $S = 110.33$ K, $Pr = 0.72$, $\gamma = 1.4$; spatial formulation at fixed $F = 2.2 \times 10^{-4}$. The *disturbance* temperature wall condition is isothermal ($\hat{T}|_w = 0$), which the authors state they used for the Fig. 15 neutral curve. The spatial sweeps use Chebyshev collocation at $N = 130$ (131 nodes), $y_{\max} = 40 \delta^*/L^*$, L^* scaling, and $\lambda/\mu = 1.2$.

The spatial QEP is run in fixed-frequency sweeps: real ω along each $F = \text{const}$ path in R returns the complex wavenumber, and the neutral curve is the $\sigma = -\alpha_i = 0$ crossing set; the first-mode branches use the same route with the discrete-mode selector (freestream decay + $y_{\max}$ -stationarity). Our LST code thus traces the full neutral curve $\omega(R)$, anchored against the two second-mode branch points quoted in the text at $F = 2.2 \times 10^{-4}$. Its second-mode neutral band reproduces the closed loop of Fig. 15 (nose near $R \approx 280$, upper branch $\omega \approx 0.225$, lower branch $\omega \approx 0.18$) across the full range. At the quoted $F = 2.2 \times 10^{-4}$, the band crosses at $R = 786.6$ (branch I; text value 806) and $R = 1009.9$ (branch II; text value 999.6). Using the adiabatic disturbance condition by mistake shifts these to 831 and 1033; the paper’s actual isothermal disturbance condition is a correction, not a fit, and tightens the match. The first-mode *cutoff* (upper) branch, re-extracted with the discrete-mode selector (freestream decay + $y_{\max}$ -stationarity, replacing the earlier phase-speed-band classifier), follows Fig. 15’s upper branch closely. The low- ω *onset* (lower) branch is *not* recovered: below $\omega \approx 0.03$ the 2-D first mode ($c_r \lesssim 0.45$) delocalises into the slow-acoustic continuous spectrum ($c_r \leq 1 - 1/M = 0.778$), leaving no clean discrete eigenmode. This limit is the same across Our LST code’s spatial and temporal collocation solvers and a QR-shooting variant, and a wave-angle sweep confirms the 2-D mode is the most unstable here (so this is a 2-D, not an oblique, first mode). Recovering the onset branch would need a compound-matrix shooting solver robust to the near-continuous-spectrum decay rates, which is left as an open item in the companion working note (Figure 17).

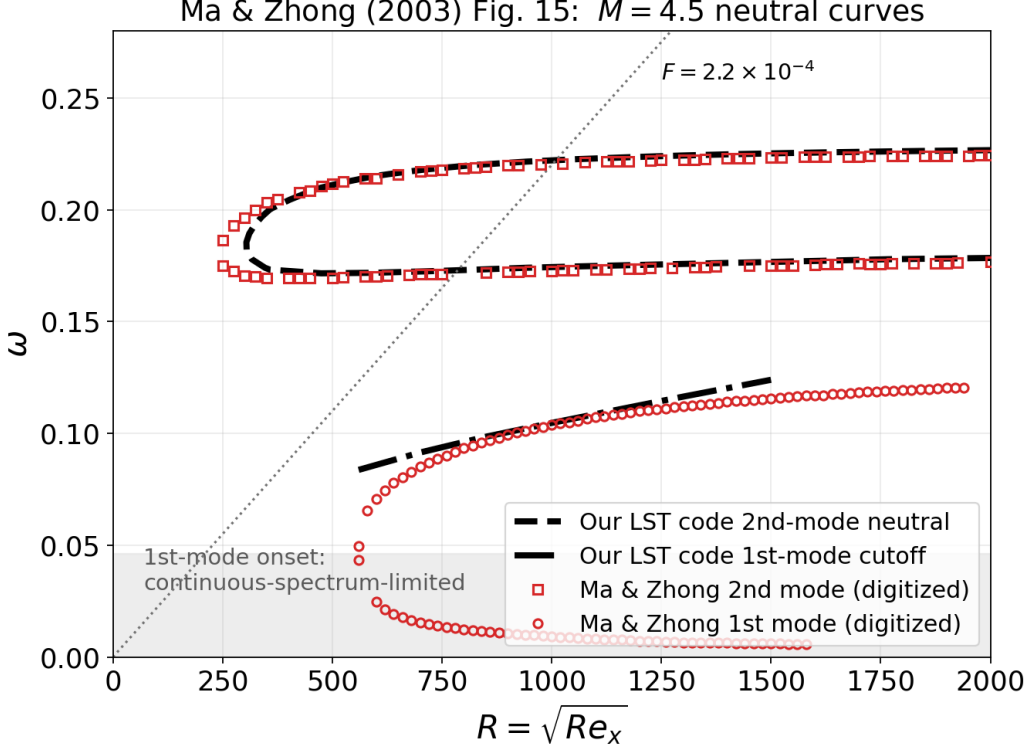


Figure 17: Ma & Zhong Fig. 15 [21], $M = 4.5$: Our LST code’s second-mode neutral loop over the pixel-digitised Fig. 15 points, matching the branch points at $F = 2.2 \times 10^{-4}$ ($R = 786.6$ vs. 806; $R = 1009.9$ vs. 999.6) under the isothermal disturbance condition. The first-mode *cutoff* branch, re-extracted with the discrete-mode selector, follows the digitised upper branch; the first-mode *onset* (shaded band) is continuous-spectrum-limited, the long-wavelength mode delocalising into the continuous spectrum, a limit shared by all of Our LST code’s collocation and QR-shooting methods (see text).

5.5 First mode: curves and families

The first mode was Our LST code’s documented weak spot. After a reference-digitisation correction, most of the Mack first-mode figures now match well; the remaining disagreement is narrow and isolated to the $M \approx 2.2$ cases (Fig. 10.1 and Fig. 10.3 at $M = 2.2$), a genuine under-amplification and too-high critical Reynolds number at that Mach rather than a general trend, and is tracked in the companion working note. This section reports the Mack first-mode figures; the Özgen–Kırcalı neutral-curve family, whose figures carry first-mode branches alongside second-mode ones, is treated separately in Section 5.6.

5.5.1 Mack (1984) Fig. 10.1: first-mode neutral loop

Reference. Mack [1], Fig. 10.1, the innermost (complete-equations) neutral loop, re-digitised with both branches plus nose.

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$), at $M = 1.6$ and $M = 2.2$; temporal neutral loop ($c_i = 0$) in $(R, F \times 10^4)$. The (R, α) -grid solves use Chebyshev collocation at $N = 140$, $y_{\max} = 45 \delta^*/L^*$, L^* scaling, and $\lambda/\mu = 0$.

Solving the temporal GEVP over an (R, α) grid (real α at each R , returning the complex phase speed) gives the neutral loop as the $c_i = 0$ contour, with the critical Reynolds number at its nose. At $M = 1.6$ the first-mode neutral loop (its lower onset and upper cutoff branches) closely tracks Mack’s complete-equations loop after the corrected reference, with critical Reynolds number $R \approx 300$ against Mack’s ≈ 215 (Figure 18). The companion $M = 2.2$ loop shows a genuine first-mode under-amplification, with a too-high critical Reynolds number ($R \approx 400$ against Mack’s ≈ 300), and is documented in the companion working note.

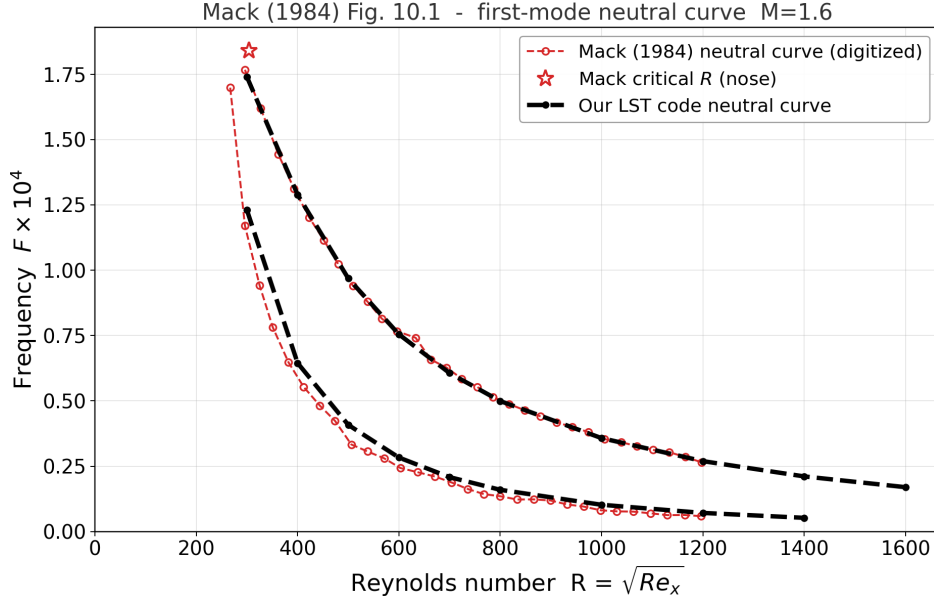


Figure 18: Mack Fig. 10.1 [1], $M = 1.6$: Our LST code’s first-mode neutral loop closely tracks Mack’s complete-equations loop, with critical Reynolds number $R \approx 300$ against Mack’s ≈ 215 .

5.5.2 Mack (1984) Fig. 10.3: oblique first-mode growth vs. R

Reference. Mack [1], Fig. 10.3.

Conditions. Air (ideal, $Pr = 0.72$), adiabatic wall; oblique waves at $M = 1.3$ ($\psi = 45^\circ$), $M = 2.2$ ($\psi = 45^\circ$), $M = 3.0$ ($\psi = 60^\circ$). These oblique cases use the exact 8×8 shooting march rather than a Chebyshev grid, with $y_{\max} \approx 4 \delta^*/L^*$ and L^* scaling.

These curves use the exact oblique first-order *shooting* route (independent of the collocation solver of Section 4, documented separately) rather than the collocation GEVP: at these Mach numbers the reduced temporal EVP’s branch selection is not trusted, so an anchor root is located by the $\sigma_{\min}(c) \rightarrow 0$ search and continued in the parameters, with the growth maximised over wavenumber at each R . At $M = 1.3$ Our LST code’s maximum first-mode temporal growth $\omega_{i,\max}(R)$ tracks Fig. 10.3 closely and reproduces the rigorous Table-10.1 eighth-order anchor; $M = 3.0$ tracks it as well. $M = 2.2$ is the exception, a genuine under-amplification (convergence-checked: not a $y_{\max}$ or step-count artifact). Topology is correct in all three (a single rising-then-turning first-mode branch), so $M = 2.2$ is an isolated outlier rather than part of a growing-with-Mach trend. The $M = 1.3$ case is shown in Figure 19; the $M = 2.2$ case is documented in the companion working note.

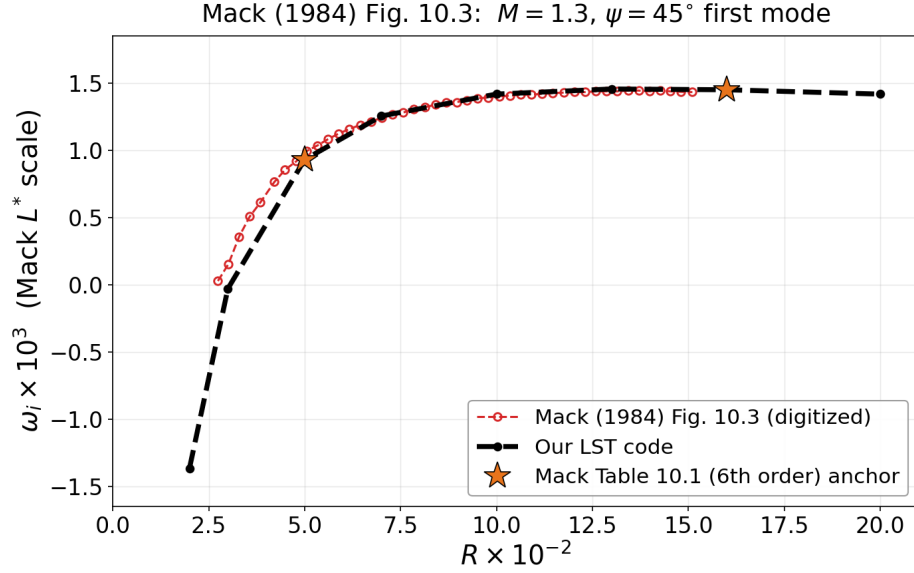


Figure 19: Mack Fig. 10.3 [1], $M = 1.3$, $\psi = 45^\circ$: Our LST code's $\omega_{i,\max}(R)$ over the digitised points, reproducing the rigorous Table-10.1 eighth-order anchor.

5.5.3 Mack (1984) Fig. 10.4: oblique first mode at high Mach

Reference. Mack [1], Fig. 10.4.

Conditions. Air, adiabatic wall; the high-Mach first mode is oblique-dominated, so growth is maximised over both α and wave angle ψ ; $M = 4.5, 5.8, 7, 10$ with $y_{\max} \approx 4\delta^*/L^*$. The 3-D temporal solves use Chebyshev collocation at $N = 200$ ($y_{\max}$ reaching $150\delta^*/L^*$ at $M = 10$), L^* scaling, and $\lambda/\mu = 0$.

The reduced 3-D temporal GEVP takes the real oblique pair α and wave angle ψ (equivalently β) and returns the complex frequency, with ω_i maximised over (α, ψ) at each R . Across $M = 4.5, 5.8, 7$, and 10 , Our LST code's maximum first-mode temporal growth $\omega_{i,\max}(R)$ tracks the digitised Fig. 10.4 points, with the oblique-first-mode peak near $\psi \approx 60^\circ$ ($c_r \approx 0.83$ at $M = 4.5$) and only a small residual at $M = 10$, the top of the Mach range. One caveat: these curves were computed with the reduced dense-EVP path rather than the exact-shooting path used for the convergence-checked Fig. 10.3 cases, and no exact-shooting cross-check was recorded; the domain used the second-mode-calibrated $\sim 4\delta^*/L^*$ rule, while the first mode reaches further from the wall. There is no first-mode under-amplification trend at high Mach in this figure. $M = 4.5$ and $M = 10$ are shown (Figures 20 and 21); $M = 5.8$ and $M = 7$ follow the same trend and are summarised in Table 1.

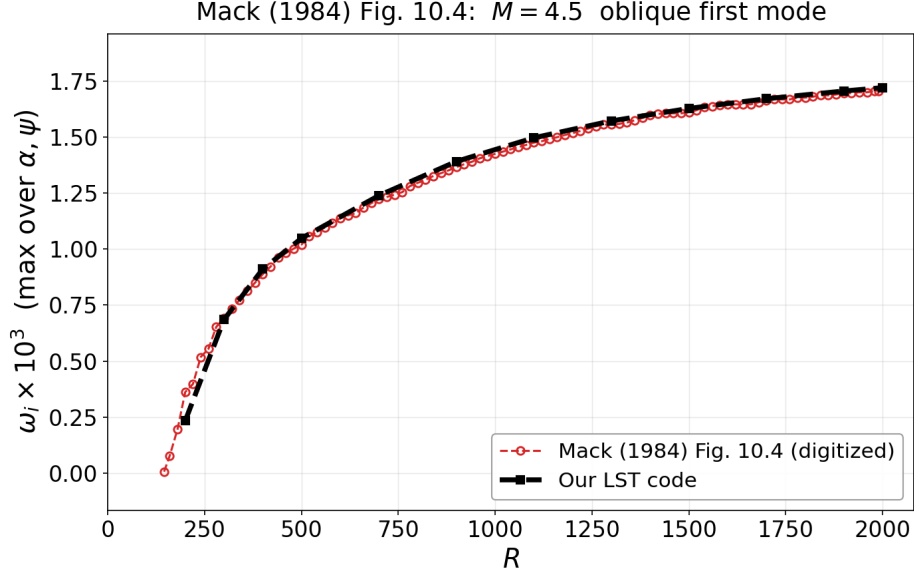


Figure 20: Mack Fig. 10.4 [1], $M = 4.5$: the oblique first mode optimised over (α, ψ) , over the digitised points.

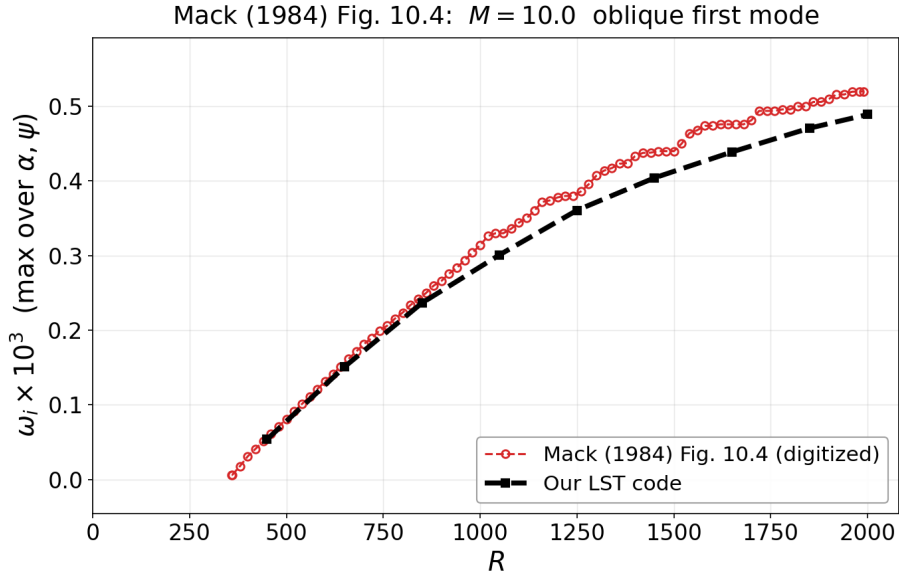


Figure 21: Mack Fig. 10.4 [1], $M = 10$: the same comparison at the top of the Mach range, with a small residual (lower-confidence magnitude, see text).

5.6 Mixed first and second mode: the Özgen–Kırcalı neutral-curve family

Fig. 3 of Özgen & Kırcalı [6] presents, for each Mach number from 2 to 10, a single neutral-curve figure in the (Re_L, α_L) plane. At $M = 2$ and 3 the figure contains the first-mode lobe alone; from $M \approx 4$ the second (Mack) mode enters, and each figure then spans *both* instability families; at $M = 6$, for example, four branches: second-mode upper and lower, first-mode cutoff, and first-mode onset. Because a single reproduced figure tests both modes at once, these cases are classified

neither under the first nor the second mode but as a mixed family, and they are reported last, drawing on both preceding sections: at $M = 2$ and $M = 3$ only the first mode is present, while $M = 4$ – 10 span both modes.

Reference. Özgen & Kırcah [6], Fig. 3 ($M = 2, 3, 4, 6, 7, 8, 10$).

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$); temporal neutral contour ($c_i = 0$) in the (Re_L, α_L) plane, with the Özgen temperature-dependent transport laws. The neutral loci are traced on two Chebyshev grids: the wall-trapped second mode at $N = 180$ with $y_{\max} = 8$ – $12 \delta^*/L^*$, and the more diffuse first mode at $N = 200$ with $y_{\max} = 35$ – $45 \delta^*/L^*$; L^* scaling throughout.

The temperature-form 2-D temporal GEVP (the Özgen equation arrangement) is solved over a (Re_L, α_L) grid, real α_L at each Re_L returning the complex phase speeds. A discrete-mode extractor (eigenfunction decay + $y_{\max}$ -stationarity, the two acceptance tests of Section 4.5) selects the physical mode at each node, eigenvalue continuation carries difficult branches through the grid, and the neutral family (all branches $\alpha_L(Re_L)$ of both the first and second mode) is the extracted $c_i = 0$ contour. This extractor replaces the earlier c_r -band classifier that had excluded the genuine first mode; with that correction and the references re-digitised, the neutral families reproduce Mach by Mach as follows.

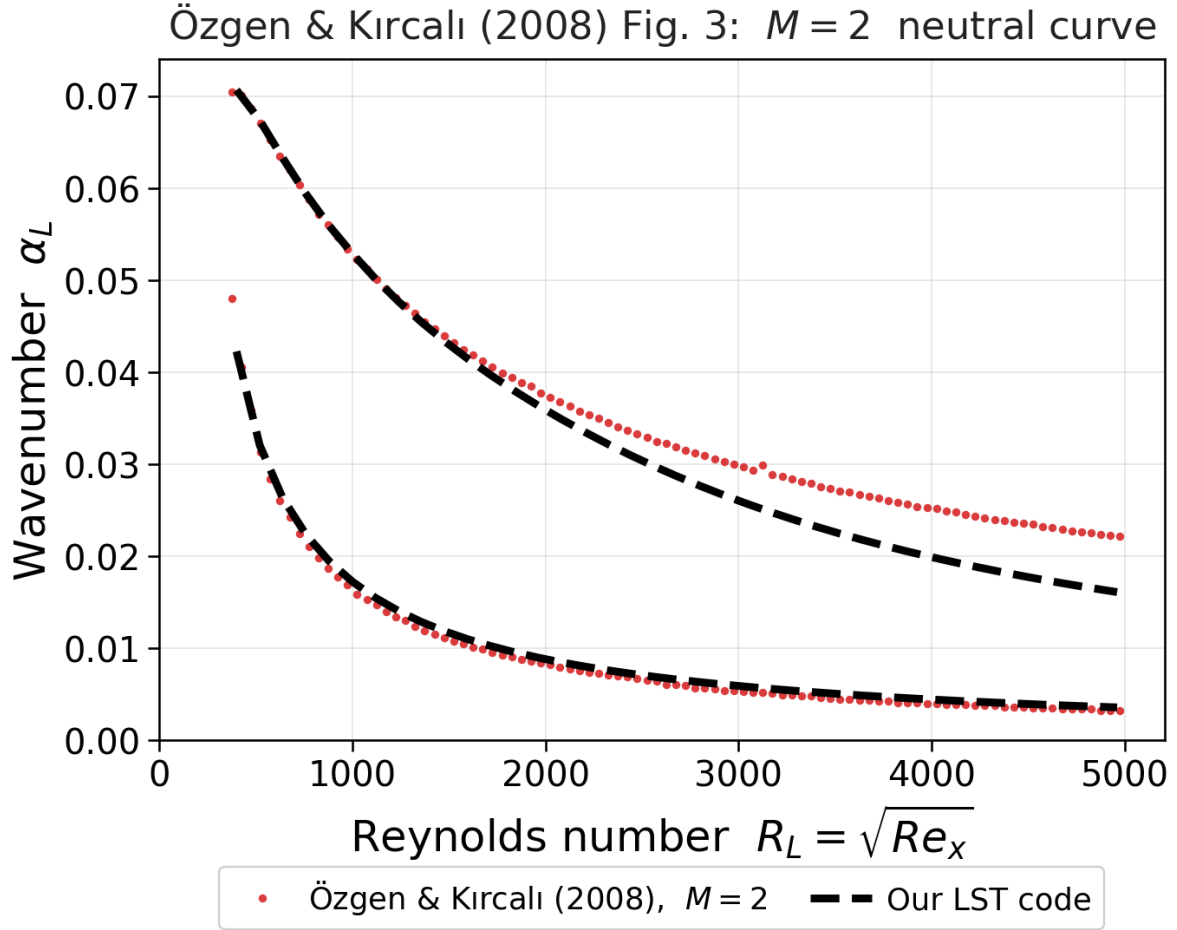


Figure 22: Özgen & Kırçalı Fig. 3 [6], $M = 2$: the first-mode lobe, the only unstable family at this Mach number.

$M = 2$ and $M = 3$: first mode only. Below the second-mode onset the figure carries a single lobe. At $M = 2$ Our LST code traces both first-mode branches over the full R -range by eigenvalue continuation (Figure 22); at $M = 3$ the cutoff branch is reproduced where resolvable, the onset being blocked by the continuous spectrum.

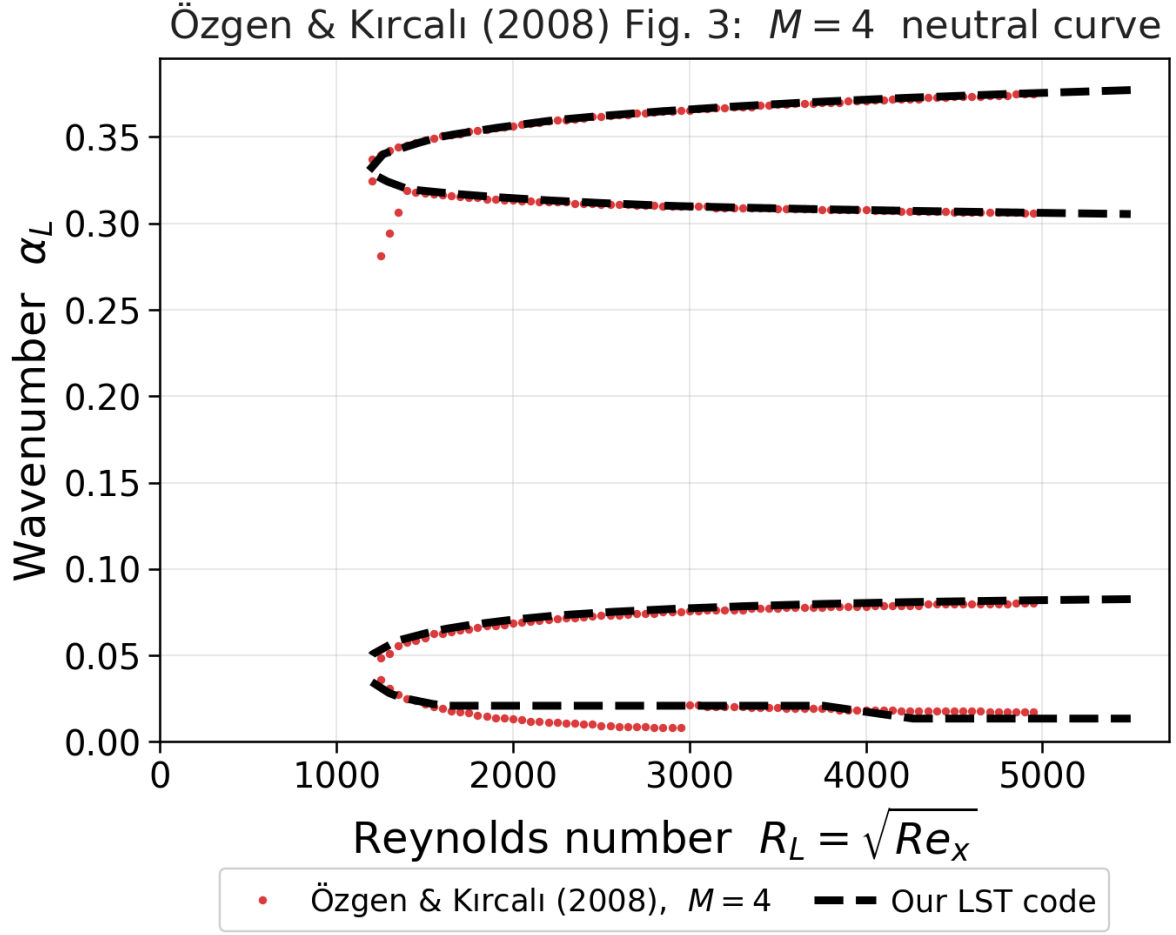


Figure 23: Özgen & Kırçalı Fig. 3 [6], $M = 4$: the second mode has entered and the figure now spans both instability families.

$M = 4$: **both modes.** From $M \approx 4$ the second (Mack) mode appears and each figure spans both families. Our LST code reproduces the entire two-lobe structure (Figure 23): both second-mode branches sit on the digitisation floor and the first-mode cutoff is closely tracked; only the deep low- α onset departs.

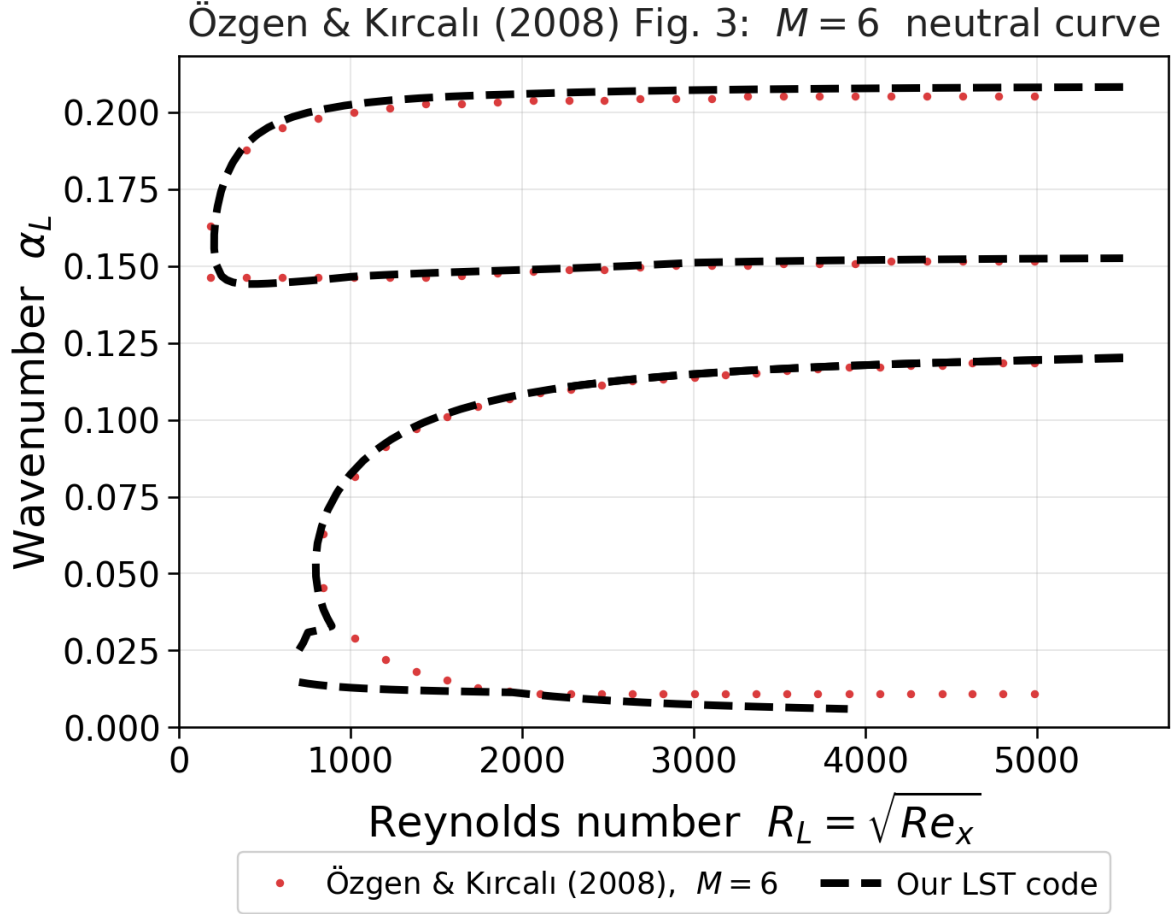


Figure 24: Özgen & Kırçalı Fig. 3 [6], $M = 6$: both modes in one figure, the re-digitised reference points against Our LST code’s full $c_i = 0$ contour.

$M = 6$: **both modes.** Here four branches coexist, second-mode upper and lower and first-mode cutoff and onset. After re-digitising the reference and re-tracing the neutral loci on a grid extended to low Re , Our LST code closes both lobes at their critical Reynolds number (second-mode nose $R \approx 206$ against Özgen’s ≈ 200) and reproduces the three robust branches (second-mode upper and lower, first-mode cutoff, Figure 24); only the first-mode onset runs below the reference, the continuous-spectrum-limited tail.

$M = 7$, $M = 8$, $M = 10$. The dominant second-mode cutoff branch is reproduced, but Our LST code predicts a *connected* first–second-mode band where the reference shows two lobes separated by a stable gap, a connected-versus-separated topology difference (inter-mode over-prediction at high Mach), not a second-mode-curve error. A separate growth-contour comparison (the $c_i = 0.004$ lobe rather than the neutral curve) likewise does not match, and the interior $c_i = 0.012$ $M = 2$ lobe is absent because Our LST code’s peak $M = 2$ growth $c_{i,\max} = 0.0109$ falls just short of 0.012. These Mach numbers ($M = 3, 7, 8, 10$) are summarised in Table 1.

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